

Chapter 14 Radiation By Moving Charges

14.1. Lienard-Wiechert Potential

Lienard-Wiechert potential is the potential for the radiation field by a moving charge. Let $\vec{r}(t)$ and $\vec{v}(t)$ be the position and velocity of a moving charge at time t in the lab frame. The charge and current density in the lab frame can be written explicitly into 3-vector form in the Euclidean space as

$$\begin{cases} \rho(\vec{x}, t) &= q \delta^{(3)}[\vec{x} - \vec{r}(t)] \\ \vec{J}(\vec{x}, t) &= q \vec{v}(t) \delta^{(3)}[\vec{x} - \vec{r}(t)] \end{cases} \quad (1)$$

To use the covariant Green function formula from Chapter 12,

$$A^\alpha(\mathbf{x}) = A_{in}^\alpha(\mathbf{x}) + \frac{4\pi}{c} \int D(\mathbf{x} - \mathbf{x}') J^\alpha(\mathbf{x}') d^4\mathbf{x}' \quad (2)$$

with the retarded Green's function

$$D(\mathbf{x} - \mathbf{x}') = \frac{1}{2\pi} \theta(x_0 - x'_0) \delta(|\mathbf{x} - \mathbf{x}'|^2) \quad (3)$$

we modify ρ and $\vec{J}$ into covariant 4-vector form.

$$\begin{aligned} \rho(\vec{x}, t) &= q \int dr_0 \delta(x_0 - r_0) \delta^{(3)}(\vec{x} - \vec{r}(t_r)) & \text{where } \begin{cases} x_0 = ct \\ r_0 = ct_r \end{cases} \\ &= q \int c dt_r \delta^{(4)}(\mathbf{x} - \mathbf{r}(t_r)) = q \int dt_r \frac{V^0}{\gamma} \delta^{(4)}(\mathbf{x} - \mathbf{r}(\tau)) \\ \vec{J}(\vec{x}, t) &= q \int dr_0 \vec{v}(t_r) \delta(x_0 - r_0) \delta^{(3)}(\vec{x} - \vec{r}(t_r)) = qc \int dt_r \frac{\vec{V}(t_r)}{\gamma} \delta^{(4)}(\mathbf{x} - \mathbf{r}(t_r)) \end{aligned}$$

where $\gamma = 1/\sqrt{1 - |\vec{v}(t_r)|^2/c^2}$ is a function of t_r and

$$\mathbf{x} = (ct, \vec{x}), \quad \mathbf{r}(t_r) = (ct_r, \vec{r}(t_r)), \quad (V^0, \vec{V}) = \gamma(c, \vec{v}(t_r)) = \gamma \dot{\mathbf{r}}(t_r)$$

are the position and velocity of the charge in the 4-dimensional Minkowski space, respectively.

The 4-vector for the charge/current density of a moving charge is thus

$$J^\alpha(\mathbf{x}) = qc \int \frac{V^\alpha(t_r)}{\gamma} \delta^{(4)}(\mathbf{x} - \mathbf{r}(t_r)) dt_r \quad (4)$$

If there is no incoming field, $A_{in}^\alpha = 0$, the potential for the radiation field is

$$\begin{aligned} A^\alpha(\mathbf{x}) &= \frac{4\pi}{c} \int d^4\mathbf{x}' D(\mathbf{x} - \mathbf{x}') J^\alpha(\mathbf{x}') \\ &= 2q \int d^4\mathbf{x}' \theta(x_0 - x'_0) \delta(|\mathbf{x} - \mathbf{x}'|^2) \int \frac{V^\alpha(t_r)}{\gamma} \delta^{(4)}(\mathbf{x}' - \mathbf{r}(t_r)) dt_r \\ &= 2q \int dt_r \frac{V^\alpha(t_r)}{\gamma} \int d^4\mathbf{x}' \delta^{(4)}(\mathbf{x}' - \mathbf{r}(t_r)) \theta(x_0 - x'_0) \delta(|\mathbf{x} - \mathbf{x}'|^2) \\ &= 2q \int \frac{dt_r}{\gamma} V^\alpha(t_r) \theta(x_0 - r_0) \delta(|\mathbf{x} - \mathbf{r}(t_r)|^2) \end{aligned} \quad (5)$$

where

$$|\mathbf{x} - \mathbf{r}(t_r)|^2 = c^2(t - t_r)^2 - |\vec{x} - \vec{r}(t_r)|^2 \quad \text{and} \quad r_0 = ct_r$$

To further evaluate the integral in Eq. (5), we use one property of δ -function,

$$\int dt_r f(t_r) \delta(g(t_r)) = \frac{f(t_r)}{|g'(t_r)|} \Big|_{g(t_r)=0}$$

For $g(t_r) = |\mathbf{x} - \mathbf{r}(t_r)|^2$,

$$\begin{aligned} \frac{d|\mathbf{x} - \mathbf{r}(t_r)|^2}{dt_r} &= -2c^2(t - t_0) + 2(\vec{x} - \vec{r}) \cdot \frac{d\vec{r}(t_r)}{dt_r} \\ &= -2\gamma^{-1} \left[(x_0 - r_0)V^0 - (\vec{x} - \vec{r}) \cdot \vec{V} \right] = -2\gamma^{-1}(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}(t_r) \end{aligned} \quad (6)$$

where, as it will be shown in the following, $(\mathbf{x} - \mathbf{r}(t_r)) \cdot \mathbf{V}(t_r) > 0$ for the retarded time t_r . The integration with $\delta(|\mathbf{x} - \mathbf{r}(t_r)|^2)$ in Eq. (5) thus yields

$$A^\alpha(\mathbf{x}) = \frac{q V^\alpha(t_r)}{[\mathbf{x} - \mathbf{r}(t_r)] \cdot \mathbf{V}(t_r)} \theta(x_0 - ct_r) \quad (7)$$

where the retarded time t_r is defined on the light cone,

$$|\mathbf{x} - \mathbf{r}(t_r)|^2 = c^2(t - t_r)^2 - |\vec{x} - \vec{r}(t_r)|^2 = 0$$

and for the retarded radiation,

$$t = t_r + \frac{1}{c} |\vec{x} - \vec{r}(t_r)| > t_r \quad \implies \quad \theta(x_0 - ct_r) = 1$$

The Lienard-Wiechert potential for the radiation of a moving charge is thus

$$A^\alpha(\mathbf{x}) = \left\{ \frac{q V^\alpha(t_r)}{[\mathbf{x} - \mathbf{r}(t_r)] \cdot \mathbf{V}(t_r)} \right\}_{ret} \quad (8)$$

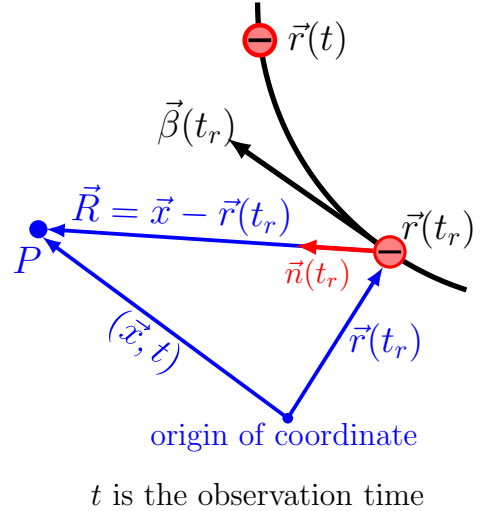
where t_r is the retarded radiation time that has to be solved from

$$ct = ct_r + |\vec{x} - \vec{r}(t_r)| \quad (9)$$

Let $\vec{R} = \vec{x} - \vec{r}(t_r)$, $R = |\vec{x} - \vec{r}(t_r)|$, and $\vec{n} = \vec{R}/R$.

$$\begin{aligned} [\mathbf{x} - \mathbf{r}(t_r)] \cdot \mathbf{V}(t_r) &= V_0(ct - ct_r) - \vec{V} \cdot [\vec{x} - \vec{r}(t_r)] \\ &= (V_0 - \vec{V} \cdot \vec{n})R = \gamma c R (1 - \vec{\beta} \cdot \vec{n}) \end{aligned}$$

Note that $(1 - \vec{\beta} \cdot \vec{n}) > 0$ which satisfies the requirement of $|g'(t_r)| > 0$ in Eq. (6) for the evaluation of the integral in Eq. (5). The Lienard-Wiechert potential for the radiation of a



moving charge can also be written in 3-vector form as

$$\Phi(\vec{x}, t) = \left\{ \frac{q}{R(t_r) [1 - \vec{\beta}(t_r) \cdot \vec{n}(t_r)]} \right\}_{ret} \quad (10)$$

$$\vec{A}(\vec{x}, t) = \left\{ \vec{\beta}(t_r) \Phi(\vec{x}, t) \right\}_{ret} \quad (11)$$

where $\vec{R} = \vec{x} - \vec{r}(t_r)$, $\vec{n} = \vec{R}/R$, and the retarded radiation time t_r as a function of t and $\vec{x}$ has to be solved from

$$ct = ct_r + |\vec{x} - \vec{r}(t_r)|$$

Radiation from a Distribution of Moving Charges

Consider an ensemble of moving charged particles. Let $f(\vec{r}, \vec{p}, t)$ be the particle distribution in phase space where $\vec{p} = \gamma m \vec{u}$ is the canonical momentum of particles and the charge density in configurational space is

$$\rho(\vec{r}, t) = \int_{-\infty}^{\infty} f(\vec{r}, \vec{p}, t) d^3\vec{p} \quad (12)$$

With the charge density, the current density can be calculated from

$$\vec{J}(\vec{r}, t) = \frac{1}{m} \int \frac{\vec{p}}{\gamma} f(\vec{r}, \vec{p}, t) d^3\vec{p} = u(\vec{r}, t) \rho(\vec{r}, t) \quad (13)$$

where $u(\vec{r}, t)$ is the average velocity in momentum space of the particles in infinitesimal volume element at $\vec{r}$ at time t .

It is difficult to directly use the Lienard-Wiechert potential in Eqs. (10) and (11) to calculate the radiation because the retardation in $\vec{\beta}(t_r)$ makes the integration of $\vec{p}$ tangled with the integration of $\vec{r}$. We thus use the original 3-dimensional retarded Green's function,

$$G(|\vec{r} - \vec{r}'|, t - t') = \frac{1}{|\vec{r} - \vec{r}'|} \delta(t' - t + |\vec{r} - \vec{r}'|/c) \quad (14)$$

The scalar and vector potential can be calculated from the 3D Green's function using

$$\begin{aligned} \Phi(\vec{r}, t) &= \int G(|\vec{r} - \vec{r}'|, t - t') \rho(\vec{r}', t') d^3\vec{r}' dt' \\ &= \int \frac{1}{|\vec{r} - \vec{r}'|} \rho(\vec{r}', t - |\vec{r} - \vec{r}'|/c) d^3\vec{r}' \end{aligned} \quad (15)$$

$$\vec{A}(\vec{r}, t) = \int \frac{\vec{u}(\vec{r}', t - |\vec{r} - \vec{r}'|/c)}{|\vec{r} - \vec{r}'|} \rho(\vec{r}', t - |\vec{r} - \vec{r}'|/c) d^3\vec{r}' \quad (16)$$

This problem is workable if the t -dependence of the charge distribution is known.

14.2. Radiation Field

$$F^{\alpha\beta} = \frac{\partial A^\beta}{\partial x_\alpha} - \frac{\partial A^\alpha}{\partial x_\beta} \quad (17)$$

$F^{\alpha\beta}$ can be calculated directly from Eq. (8), but the derivatives involving the retardation condition is cumbersome. To avoid the derivative with the retardation, we use the integral form of A^α in Eq. (5).

$$\begin{aligned} \frac{\partial A^\beta(\mathbf{x})}{\partial x_\alpha} &= 2q \int dt_r \gamma^{-1} V^\beta(t_r) \frac{\partial}{\partial x_\alpha} [\theta(x_0 - r_0) \delta(|\mathbf{x} - \mathbf{r}(t_r)|^2)] \\ &= 2q \int dt_r \gamma^{-1} V^\beta(t_r) \theta(x_0 - r_0) \frac{\partial}{\partial x_\alpha} \delta(|\mathbf{x} - \mathbf{r}(t_r)|^2) \end{aligned}$$

where

$$\delta(|\mathbf{x} - \mathbf{r}(t_r)|^2) \frac{\partial}{\partial x_\alpha} \theta(x_0 - r_0) = \begin{cases} 0 & \text{for } \alpha \neq 0 \\ \delta(|\mathbf{x} - \mathbf{r}(t_r)|^2) \delta(x_0 - r_0) = 0 & \text{for } \alpha = 0 \end{cases}$$

because the two δ -functions in the case of $\alpha = 0$ contradict each other except the trivial case of $\mathbf{x} = \mathbf{r}(t_r)$ which is excluded from the consideration of radiation. In order to utilize the derivative of δ -function in integral, we switch the derivative as

$$\frac{\partial \delta(f(\mathbf{x}, t_r))}{\partial x_\alpha} = \frac{d\delta(f)}{df} \frac{\partial f}{\partial x_\alpha} = \frac{\partial f}{\partial x_\alpha} \left(\frac{\partial f}{\partial t_r} \right)^{-1} \frac{\partial \delta(f(\mathbf{x}, t_r))}{\partial t_r}$$

where

$$f(\mathbf{x}, t_r) = |\mathbf{x} - \mathbf{r}(t_r)|^2 = (x_0 - ct_r)^2 - |\vec{x} - \vec{r}(t_r)|^2$$

and

$$\begin{aligned} \frac{\partial f(\mathbf{x}, t_r)}{\partial x_\alpha} &= 2(\mathbf{x} - \mathbf{r})^\alpha \\ \frac{\partial f(\mathbf{x}, t_r)}{\partial t_r} &= -2(\mathbf{x} - \mathbf{r})^\beta \frac{dr_\beta(t_r)}{dt_r} = -\frac{2(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}}{\gamma} \\ \frac{\partial}{\partial x_\alpha} \delta(|\mathbf{x} - \mathbf{r}|^2) &= -\frac{\gamma(\mathbf{x} - \mathbf{r})^\alpha}{(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}} \frac{\partial}{\partial t_r} \delta(|\mathbf{x} - \mathbf{r}|^2) \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{\partial A^\beta}{\partial x_\alpha} &= -2q \int dt_r V^\beta \theta(x_0 - ct_r) \frac{(\mathbf{x} - \mathbf{r})^\alpha}{(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}} \frac{\partial}{\partial t_r} \delta(|\mathbf{x} - \mathbf{r}|^2) \\ &= 2q \int dt_r \theta(x_0 - ct_r) \delta(|\mathbf{x} - \mathbf{r}|^2) \frac{\partial}{\partial t_r} \frac{(\mathbf{x} - \mathbf{r})^\alpha V^\beta}{(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}} \\ &\quad + 2qc \int dt_r \delta(x_0 - ct_r) \delta(|\mathbf{x} - \mathbf{r}|^2) \frac{(\mathbf{x} - \mathbf{r})^\alpha V^\beta}{(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}} \end{aligned}$$

The second integral is zero because again the two δ -functions contradict each other except the trivial case of $\mathbf{x} = \mathbf{r}(t_r)$.

$$\begin{aligned}\frac{\partial A^\beta}{\partial x_\alpha} &= 2q \left\{ \left| \frac{\partial |\mathbf{x} - \mathbf{r}(t_r)|^2}{\partial t_r} \right|^{-1} \frac{\partial}{\partial t_r} \frac{[\mathbf{x} - \mathbf{r}(t_r)]^\alpha V^\beta(t_r)}{[\mathbf{x} - \mathbf{r}(t_r)] \cdot \mathbf{V}(t_r)} \right\}_{r_0=x_0-|\vec{x}-\vec{r}(t_r)|} \\ &= \left\{ \frac{q\gamma}{|(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}|} \frac{\partial}{\partial t_r} \frac{[\mathbf{x} - \mathbf{r}(t_r)]^\alpha V^\beta(t_r)}{[\mathbf{x} - \mathbf{r}(t_r)] \cdot \mathbf{V}(t_r)} \right\}_{r_0=x_0-|\vec{x}-\vec{r}(t_r)|}\end{aligned}$$

where $(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V} > 0$ as shown in the following. The radiation field is then

$$\begin{aligned}F^{\alpha\beta}(\mathbf{x}) &= \frac{q\gamma}{(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}} \frac{\partial}{\partial t_r} \left[\frac{(\mathbf{x} - \mathbf{r})^\alpha V^\beta - (\mathbf{x} - \mathbf{r})^\beta V^\alpha}{(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}} \right] \\ &= \frac{q\gamma}{[(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}]^2} \frac{\partial}{\partial t_r} [(\mathbf{x} - \mathbf{r})^\alpha V^\beta - (\mathbf{x} - \mathbf{r})^\beta V^\alpha] \\ &\quad - \frac{q\gamma}{[(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}]^3} [(\mathbf{x} - \mathbf{r})^\alpha V^\beta - (\mathbf{x} - \mathbf{r})^\beta V^\alpha] \frac{\partial}{\partial t_r} [(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}] \quad (18)\end{aligned}$$

where $\mathbf{r}$, $\mathbf{V}$ and γ are functions of t_r and the retarded radiation time t_r has to be solved from

$$ct = ct_r - |\vec{x} - \vec{r}(t_r)|$$

To further simplify $F^{\alpha\beta}$, let

$$\begin{aligned}R &= |\vec{x} - \vec{r}| = x_0 - r_0, & \vec{n} &= \frac{1}{R}(\vec{x} - \vec{r}), \\ \mathbf{C}_1 &= (1, \vec{n}), & \mathbf{C}_2 &= (1, \vec{\beta}), & \mathbf{C}_3 &= (0, \dot{\vec{\beta}})\end{aligned}$$

where

$$\mathbf{C}_1 \cdot \mathbf{C}_2 = 1 - \vec{n} \cdot \vec{\beta} \quad \text{and} \quad \mathbf{C}_1 \cdot \mathbf{C}_3 = -\vec{n} \cdot \dot{\vec{\beta}}$$

Then

$$\begin{aligned}\mathbf{x} - \mathbf{r} &= (x_0 - r_0, \vec{x} - \vec{r}) = R(1, \vec{n}) = R\mathbf{C}_1 \\ \mathbf{V} &= (\gamma c, \gamma c \vec{\beta}) = \gamma c(1, \vec{\beta}) = \gamma c \mathbf{C}_2 \\ (\mathbf{x} - \mathbf{r}) \cdot \mathbf{V} &= R\gamma c \mathbf{C}_1 \cdot \mathbf{C}_2 = R\gamma c(1 - \vec{n} \cdot \vec{\beta}) \\ \frac{\partial}{\partial t_r}(\mathbf{x} - \mathbf{r}) &= -\frac{d\mathbf{r}}{dt_r} = -\frac{1}{\gamma}\mathbf{V} = -c\mathbf{C}_2\end{aligned}$$

$$\begin{aligned}\frac{d\mathbf{V}}{dt_r} &= \left(c \frac{d\gamma}{dt_r}, c \frac{d\gamma}{dt_r} \vec{\beta} + c\gamma \frac{d\vec{\beta}}{dt_r} \right) \\ &= \left(c\gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}}), c\gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}})\vec{\beta} + c\gamma \dot{\vec{\beta}} \right) = c\gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}})\mathbf{C}_2 + c\gamma \mathbf{C}_3\end{aligned}$$

where

$$\dot{\vec{\beta}} = \frac{d\vec{\beta}}{dt_r} \quad \text{and} \quad \frac{d\gamma}{dt_r} = \gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}})$$

$$\begin{aligned}
\frac{\partial}{\partial t_r} [(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}] &= -\frac{1}{\gamma} \mathbf{V} \cdot \mathbf{V} + (\mathbf{x} - \mathbf{r}) \cdot \frac{d\mathbf{V}}{dt_r} \\
&= -\frac{c^2}{\gamma} + Rc\gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}}) \mathbf{C}_1 \cdot \mathbf{C}_2 + Rc\gamma \mathbf{C}_1 \cdot \mathbf{C}_3 \\
&= -\frac{c^2}{\gamma} + Rc\gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}}) (1 - \vec{n} \cdot \vec{\beta}) - Rc\gamma \vec{n} \cdot \dot{\vec{\beta}} \\
\frac{\partial}{\partial t_r} (\mathbf{x} - \mathbf{r})^\alpha V^\beta &= -\frac{1}{\gamma} V^\alpha V^\beta + (\mathbf{x} - \mathbf{r})^\alpha \frac{dV^\beta}{dt_r} \\
\frac{\partial}{\partial t_r} [(\mathbf{x} - \mathbf{r})^\alpha V^\beta - (\mathbf{x} - \mathbf{r})^\beta V^\alpha] &= (\mathbf{x} - \mathbf{r})^\alpha \frac{dV^\beta}{dt_r} - (\mathbf{x} - \mathbf{r})^\beta \frac{dV^\alpha}{dt_r} \\
&= Rc\gamma^3(\vec{\beta} \cdot \dot{\vec{\beta}}) \left(\mathbf{C}_1^\alpha \mathbf{C}_2^\beta - \mathbf{C}_1^\beta \mathbf{C}_2^\alpha \right) + Rc\gamma \left(\mathbf{C}_1^\alpha \mathbf{C}_3^\beta - \mathbf{C}_1^\beta \mathbf{C}_3^\alpha \right) \\
(\mathbf{x} - \mathbf{r})^\alpha V^\beta - (\mathbf{x} - \mathbf{r})^\beta V^\alpha &= R\gamma c \left(\mathbf{C}_1^\alpha \mathbf{C}_2^\beta - \mathbf{C}_1^\beta \mathbf{C}_2^\alpha \right)
\end{aligned}$$

Thus,

$$\begin{aligned}
F^{\alpha\beta}(\mathbf{x}) &= \frac{q}{\gamma c^2 R^2 (1 - \vec{n} \cdot \vec{\beta})^2} \frac{\partial}{\partial t_r} [(\mathbf{x} - \mathbf{r})^\alpha V^\beta - (\mathbf{x} - \mathbf{r})^\beta V^\alpha] \\
&- \frac{q}{\gamma^2 c^3 R^3 (1 - \vec{n} \cdot \vec{\beta})^3} [(\mathbf{x} - \mathbf{r})^\alpha V^\beta - (\mathbf{x} - \mathbf{r})^\beta V^\alpha] \frac{\partial}{\partial t_r} [(\mathbf{x} - \mathbf{r}) \cdot \mathbf{V}] \\
&= \frac{q}{cR(1 - \vec{n} \cdot \vec{\beta})^2} \left[\gamma^2(\vec{\beta} \cdot \dot{\vec{\beta}}) \left(\mathbf{C}_1^\alpha \mathbf{C}_2^\beta - \mathbf{C}_1^\beta \mathbf{C}_2^\alpha \right) + \left(\mathbf{C}_1^\alpha \mathbf{C}_3^\beta - \mathbf{C}_1^\beta \mathbf{C}_3^\alpha \right) \right] \\
&- \frac{q}{cR(1 - \vec{n} \cdot \vec{\beta})^3} \left(\mathbf{C}_1^\alpha \mathbf{C}_2^\beta - \mathbf{C}_1^\beta \mathbf{C}_2^\alpha \right) \left[\gamma^2(\vec{\beta} \cdot \dot{\vec{\beta}}) (1 - \vec{n} \cdot \vec{\beta}) - \vec{n} \cdot \dot{\vec{\beta}} - \frac{c}{R\gamma^2} \right] \\
&= \frac{q}{R(1 - \vec{n} \cdot \vec{\beta})^3} \left\{ \left(\frac{1}{R\gamma^2} + \frac{1}{c} \vec{n} \cdot \dot{\vec{\beta}} \right) \left(\mathbf{C}_1^\alpha \mathbf{C}_2^\beta - \mathbf{C}_1^\beta \mathbf{C}_2^\alpha \right) \right. \\
&\quad \left. + \frac{1}{c} (1 - \vec{n} \cdot \vec{\beta}) \left(\mathbf{C}_1^\alpha \mathbf{C}_3^\beta - \mathbf{C}_1^\beta \mathbf{C}_3^\alpha \right) \right\} \tag{19}
\end{aligned}$$

For the electric field, the calculation of $E_i = F^{i0}$ needs

$$\mathbf{C}_1^i \mathbf{C}_2^0 - \mathbf{C}_1^0 \mathbf{C}_2^i = \vec{n} - \vec{\beta} \quad \text{and} \quad \mathbf{C}_1^i \mathbf{C}_3^0 - \mathbf{C}_1^0 \mathbf{C}_3^i = -\dot{\vec{\beta}}$$

Thus

$$\begin{aligned}
\vec{E} &= \frac{q}{R(1 - \vec{n} \cdot \vec{\beta})^3} \left\{ \left(\frac{1}{R\gamma^2} + \frac{1}{c} \vec{n} \cdot \dot{\vec{\beta}} \right) (\vec{n} - \vec{\beta}) - \frac{1}{c} (1 - \vec{n} \cdot \vec{\beta}) \dot{\vec{\beta}} \right\} \\
&= \frac{q(\vec{n} - \vec{\beta})}{\gamma^2(1 - \vec{n} \cdot \vec{\beta})^3 R^2} + \frac{q}{c(1 - \vec{n} \cdot \vec{\beta})^3 R} [(\vec{n} \cdot \dot{\vec{\beta}})(\vec{n} - \vec{\beta}) - \vec{n} \cdot (\vec{n} - \vec{\beta}) \dot{\vec{\beta}}]
\end{aligned}$$

where

$$[(\vec{n} \cdot \dot{\vec{\beta}})(\vec{n} - \vec{\beta}) - \vec{n} \cdot (\vec{n} - \vec{\beta}) \dot{\vec{\beta}}] = \vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]$$

Similarly, the magnetic field can be calculated from F^{ij} . The final form of the electromagnetic field is

$$\vec{E}(\vec{x}, t) = q \left\{ \frac{\vec{n} - \vec{\beta}}{\gamma^2 R^2 (1 - \vec{n} \cdot \vec{\beta})^3} + \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{c R (1 - \vec{n} \cdot \vec{\beta})^3} \right\}_{ret} \quad (20)$$

$$\vec{B}(\vec{x}, t) = (\vec{n} \times \vec{E})_{ret} \quad (21)$$

where $\vec{R} = \vec{x} - \vec{r}(t_r)$, $\vec{n} = \vec{R}/R$, $R = |\vec{R}|$, $\vec{\beta} = \vec{v}/c$, and γ are functions of the retarded time t_r that has to be solved from

$$ct = ct_r - |\vec{x} - \vec{r}(t_r)|$$

The first term in the radiation field in Eq. (20) is independent of the acceleration of the charge and is, therefore, the static field,

$$\vec{E}_{static} = q \left[\frac{\vec{n} - \vec{\beta}}{\gamma^2 R^2 (1 - \vec{\beta} \cdot \vec{n})^3} \right] \sim \frac{1}{R^2}$$

The second term in Eq. (20) depends on $\dot{\vec{\beta}}$ and is the radiation field,

$$\vec{E}_{rad} = q \left\{ \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{c R (1 - \vec{\beta} \cdot \vec{n})^3} \right\}_{ret} \sim \frac{1}{R}$$

Note that $\vec{n}$ is the direction of the wave propagation, and

$$\vec{n} \cdot \vec{E}_{rad} = 0 \quad \text{and} \quad \vec{n} \cdot \vec{B}_{rad} = 0$$

t is the observation time

The radiation field is therefore polarized on the transverse plane and decays with $1/R$. The $1/R$ decay guarantees the conservation of the total radiation energy on the wave front.

Example 1. Radiation far away ($|\vec{x}| \gg L$) from an electron oscillating inside an antenna. The motion of the electron can be written as

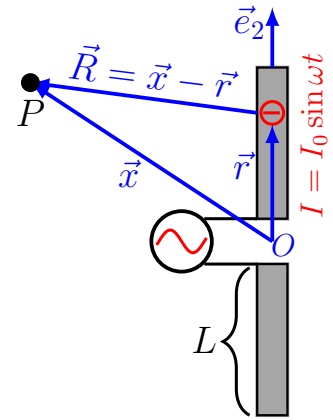
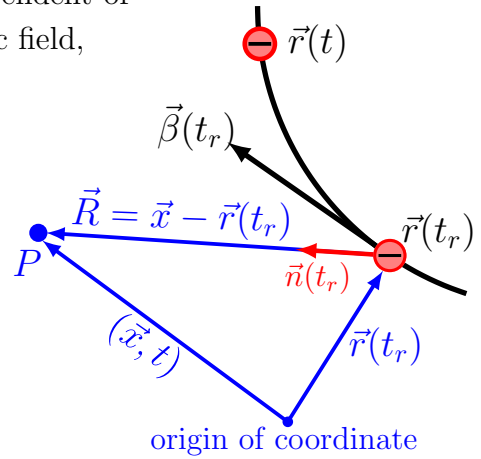
$$\begin{cases} \vec{r}(t) = (L \sin \omega t) \vec{e}_2 \\ \vec{v}(t) = (\omega L \cos \omega t) \vec{e}_2 \\ \dot{\vec{v}}(t) = -(\omega^2 L \sin \omega t) \vec{e}_2 \end{cases} \quad \text{where} \quad |\vec{v}| \ll c$$

where L is one half length of the antenna. The observation point P is at

$$\vec{x} = x_1 \vec{e}_1 + x_2 \vec{e}_2 + x_3 \vec{e}_3 \quad \text{where} \quad |\vec{x}| \gg |\vec{r}|$$

With $|\vec{x}| \gg |\vec{r}|$, the retardation condition

$$c(t - t_r) = |\vec{x} - \vec{r}(t_r)| \simeq |\vec{x}|$$



yields

$$t_r = t + |\vec{x}|/c, \quad R = |\vec{x} - \vec{r}(t_r)| \simeq |\vec{x}|, \quad \vec{n} = \frac{\vec{x} - \vec{r}(t_r)}{R} \simeq \frac{\vec{x}}{|\vec{x}|}$$

and

$$\begin{aligned} \vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}] &= \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) = (\vec{n} \cdot \dot{\vec{\beta}})\vec{n} - (\vec{n} \cdot \vec{n})\dot{\vec{\beta}} = \frac{1}{|\vec{x}|^2}(\vec{x} \cdot \dot{\vec{\beta}})\vec{x} - \dot{\vec{\beta}} \\ &= -\frac{1}{c|\vec{x}|^2}(x_2\omega^2 L \sin \omega t_r)(x_1\vec{e}_1 + x_2\vec{e}_2 + x_3\vec{e}_3) + \frac{1}{c}(\omega^2 L \sin \omega t_r)\vec{e}_2 \\ &= \frac{\omega^2 L \sin \omega t_r}{c} \left[\left(1 - \frac{x_2^2}{|\vec{x}|^2}\right) \vec{e}_2 - \frac{x_2}{|\vec{x}|^2}(x_1\vec{e}_1 + x_3\vec{e}_3) \right] \end{aligned}$$

Therefore,

$$\vec{E}_{rad}(\vec{x}, t) \simeq \frac{qLk^2}{|\vec{x}|} \left[\left(1 - \frac{x_2^2}{|\vec{x}|^2}\right) \vec{e}_2 - \frac{x_2}{|\vec{x}|^2}(x_1\vec{e}_1 + x_3\vec{e}_3) \right] \sin(\omega t - k|\vec{x}|)$$

where $k = \omega/c$. Note that $\vec{E}_{rad}$ is transverse wave because

$$\vec{n} \cdot \vec{E}_{rad} \sim (x_1\vec{e}_1 + x_2\vec{e}_2 + x_3\vec{e}_3) \cdot \left[\left(1 - \frac{x_2^2}{|\vec{x}|^2}\right) \vec{e}_2 - \frac{x_2}{|\vec{x}|^2}(x_1\vec{e}_1 + x_3\vec{e}_3) \right] = 0$$

and

$$E_{rad}(\vec{x}, t) = \frac{qLk^2}{|\vec{x}|} \left(1 - \frac{x_2^2}{|\vec{x}|^2}\right)^{1/2} \sin(\omega t - k|\vec{x}|)$$

where $E(\vec{x}, t) = 0$ when $\vec{x}$ is parallel to the antenna ($|\vec{x}| = x_2$) and the radiation field is rotational symmetric around the antenna.

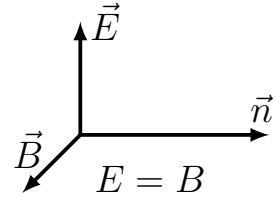
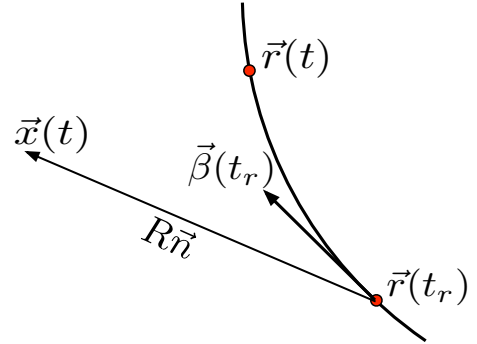
14.3. Power Radiated by an Accelerated Charge

The radiation field from an accelerated charge is

$$\begin{aligned} \vec{E}(\vec{x}, t) &= \frac{e}{c} \left\{ \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{R(1 - \vec{\beta} \cdot \vec{n})^3} \right\}_{retard} \\ \vec{B}(\vec{x}, t) &= \vec{n} \times \vec{E} \end{aligned}$$

where $R\vec{n} = \vec{x} - \vec{r}(t_r)$ and the retardation condition is $c(t - t_r) = R$. Note that $\vec{n}$ is the direction of the wave and $\vec{E} \perp \vec{B} \perp \vec{n}$. The Poynting vector is

$$\begin{aligned} \vec{S}(\vec{x}, t) &= \frac{c}{4\pi} (\vec{E} \times \vec{B}) = \frac{c}{4\pi} [\vec{E} \times (\vec{n} \times \vec{E})] \\ &= \frac{c}{4\pi} [|\vec{E}|^2 \vec{n} - (\vec{n} \cdot \vec{E}) \vec{E}] \\ &= \frac{c}{4\pi} |\vec{E}|^2 \vec{n} = \frac{c}{4\pi} |\vec{B}|^2 \vec{n} \end{aligned}$$



which is the energy transfer (flux) per unit area per unit time of an electromagnetic field.

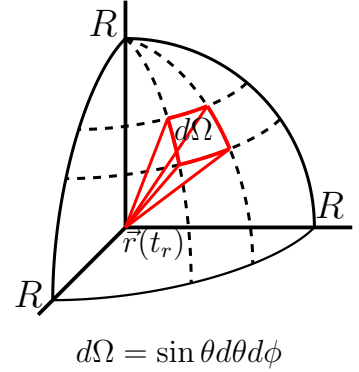
The power radiated per unit solid angle observed at location $\vec{R}$ and time t is then

$$dP(\vec{x}, t) = (\vec{S} \cdot \vec{n}) R^2 d\Omega$$

where the infinitesimal surface area at $\vec{R}$ on the solid angle $d\Omega$ is

$$R d\theta \times R \sin \theta d\phi = R^2 \sin \theta d\theta d\phi = R^2 d\Omega$$

Therefore, the radiation power per unit solid angle observed at location $\vec{x}$ and time t is



$$\frac{dP(\vec{x}, t)}{d\Omega} = \frac{c}{4\pi} R^2 |\vec{E}|^2 = \frac{c}{4\pi} R^2 |\vec{B}|^2 = \frac{e^2}{4\pi c} \left| \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{\beta} \cdot \vec{n})^3} \right|_{\text{retard}}^2 \quad (22)$$

where $\vec{n}$ is the unit vector in the direction of the solid angle,

$$\vec{n} = \frac{\vec{x} - \vec{r}(t_r)}{|\vec{x} - \vec{r}(t_r)|} = \sin \theta \cos \phi \vec{e}_x + \sin \theta \sin \phi \vec{e}_y + \cos \theta \vec{e}_z \quad (23)$$

In Eq. (22), $\vec{n} = \vec{n}(t_r)$, $\vec{\beta} = \vec{\beta}(t_r)$, and $\dot{\vec{\beta}} = \dot{\vec{\beta}}(t_r)$ are functions of t_r which, in turn, depends on the observation time t through the retardation condition,

$$t = t_r + |\vec{x} - \vec{r}(t_r)|/c$$

The radiation energy per unit solid angle can be calculated from the radiation power using

$$\mathcal{E} = \int \frac{dP(\vec{x}, t)}{d\Omega} dt \quad (24)$$

where the integration over the observation time t is tangled with the solid-angle variables (θ, ϕ) due to the inexplicit t -dependence of $\vec{n}$, $\vec{\beta}$, and $\dot{\vec{\beta}}$. In order to eliminate the complication of the retardation condition, the retarded time t_r , instead of the observation time t , is preferred as the time variable for the power. The retarded time t_r is also directly related to the position of the charge in motion. To transfer t to t_r in the integral in Eq. (24), from the retardation condition, we have

$$\frac{dt}{dt_r} = 1 + \frac{1}{c} \frac{d|\vec{x} - \vec{r}(t_r)|}{dt_r} = 1 - \frac{\vec{x} - \vec{r}(t_r)}{c|\vec{x} - \vec{r}(t_r)|} \cdot \frac{d\vec{r}(t_r)}{dt_r} = 1 - \vec{\beta} \cdot \vec{n}$$

The radiation energy per unit solid angle can be calculated from the integration over t_r as

$$\mathcal{E} = \int \frac{dP(\vec{x}, t)}{d\Omega} \left(\frac{dt}{dt_r} \right) dt_r = \int \frac{dP(t_r)}{d\Omega} dt_r \quad (25)$$

where the radiation power per solid angle in terms of the retarded time t_r is

$$\frac{dP(t_r)}{d\Omega} = \frac{dP(\vec{x}, t)}{d\Omega} \left(\frac{dt}{dt_r} \right) = \frac{e^2}{4\pi c} \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{\beta} \cdot \vec{n})^5} \quad (26)$$

Note that for $P(t_r)$ in Eqs. (26), $\vec{n}$ is an arbitrary unit vector for an arbitrary direction of the radiation and independent of t_r , while $\vec{\beta} = \vec{\beta}(t_r)$ $\dot{\vec{\beta}} = \dot{\vec{\beta}}(t_r)$ are explicit functions of t_r from the motion of the charge which gives the t_r -dependence of $P(t_r)$. The total radiation power radiated to all directions at a given time during the particle motion is

$$P(t_r) = \frac{q^2}{4\pi c} \int \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{\beta} \cdot \vec{n})^5} d\Omega \quad (27)$$

where

$$\vec{n} = \sin \theta \cos \phi \vec{e}_x + \sin \theta \sin \phi \vec{e}_y + \cos \theta \vec{e}_z$$

and

$$\begin{aligned} |\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2 &= |(\vec{n} \cdot \dot{\vec{\beta}})(\vec{n} - \vec{\beta}) - (1 - \vec{n} \cdot \vec{\beta})\dot{\vec{\beta}}|^2 \\ &= (\vec{n} \cdot \dot{\vec{\beta}})^2(1 + \beta^2 - 2\vec{n} \cdot \vec{\beta}) + (1 - \vec{n} \cdot \vec{\beta})^2\dot{\beta}^2 - 2(1 - \vec{n} \cdot \vec{\beta}) [(\vec{n} \cdot \dot{\vec{\beta}})^2 - (\vec{n} \cdot \dot{\vec{\beta}})(\vec{\beta} \cdot \dot{\vec{\beta}})] \\ &= (\vec{n} \cdot \dot{\vec{\beta}})^2(\beta^2 - 1) + (1 - \vec{n} \cdot \vec{\beta})^2\dot{\beta}^2 + 2(1 - \vec{n} \cdot \vec{\beta})(\vec{n} \cdot \dot{\vec{\beta}})(\vec{\beta} \cdot \dot{\vec{\beta}}) \end{aligned} \quad (28)$$

To further evaluate the integral over the solid angle for $P(t_r)$, we can choose a coordinate such that

$$\vec{\beta} = \beta \vec{e}_z \quad \text{and} \quad \dot{\vec{\beta}} = \dot{\beta}_x \vec{e}_x + \dot{\beta}_y \vec{e}_y + \dot{\beta}_z \vec{e}_z$$

The 1st term in the integral for $P(t_r)$ in Eqs. (27) and (28) is

$$\begin{aligned} &\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{(\vec{n} \cdot \dot{\vec{\beta}})^2(\beta^2 - 1)}{(1 - \vec{\beta} \cdot \vec{n})^5} \\ &= (\beta^2 - 1) \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{(\dot{\beta}_x \sin \theta \cos \phi + \dot{\beta}_y \sin \theta \sin \phi + \dot{\beta}_z \cos \theta)^2}{(1 - \beta \cos \theta)^5} \\ &= -\frac{\pi}{\gamma^2} \int_0^\pi \frac{(\dot{\beta}_x^2 + \dot{\beta}_y^2) \sin^2 \theta + 2\dot{\beta}_z^2 \cos^2 \theta}{(1 - \beta \cos \theta)^5} \sin \theta d\theta \\ &= -\frac{\pi}{\gamma^2} \left[(\dot{\beta}_x^2 + \dot{\beta}_y^2) \int_{-1}^1 \frac{dx}{(1 - \beta x)^5} - (\dot{\beta}_x^2 + \dot{\beta}_y^2 - 2\dot{\beta}_z^2) \int_{-1}^1 \frac{x^2 dx}{(1 - \beta x)^5} \right] \\ &= -\frac{\pi}{\gamma^2} \left[2\gamma^8(1 + \beta^2)(\dot{\beta}_x^2 + \dot{\beta}_y^2) - \frac{2}{3}\gamma^8(1 + 5\beta^2)(\dot{\beta}_x^2 + \dot{\beta}_y^2 - 2\dot{\beta}_z^2) \right] \\ &= -\frac{4\pi}{3}\gamma^6 \left[(1 - \beta^2)(\dot{\beta}_x^2 + \dot{\beta}_y^2) + (1 + 5\beta^2)\dot{\beta}_z^2 \right] \\ &= -\frac{4\pi}{3}\gamma^4 \left(\dot{\beta}^2 + 6\gamma^2\beta^2\dot{\beta}_z^2 \right) = -\frac{4\pi}{3}\gamma^4 \left[\dot{\beta}^2 + 6\gamma^2(\vec{\beta} \cdot \dot{\vec{\beta}})^2 \right] \end{aligned}$$

The 2nd term in the integral in Eqs. (27) and (28) is

$$\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{(1 - \vec{n} \cdot \vec{\beta})^2 \dot{\beta}^2}{(1 - \vec{\beta} \cdot \vec{n})^5} = 2\pi \dot{\beta}^2 \int_{-1}^1 \frac{dx}{(1 - \beta x)^3} = 4\pi \gamma^4 \dot{\beta}^2$$

and the 3rd term in the integral in Eqs. (27) and (28) is

$$\begin{aligned}
& \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{2(1 - \vec{n} \cdot \vec{\beta})(\vec{n} \cdot \dot{\vec{\beta}})(\vec{\beta} \cdot \dot{\vec{\beta}})}{(1 - \vec{\beta} \cdot \vec{n})^5} \\
&= 2(\vec{\beta} \cdot \dot{\vec{\beta}}) \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{\dot{\beta}_x \sin \theta \cos \phi + \dot{\beta}_y \sin \theta \sin \phi + \dot{\beta}_z \cos \theta}{(1 - \beta \cos \theta)^4} \\
&= 4\pi(\vec{\beta} \cdot \dot{\vec{\beta}})\dot{\beta}_z \int_{-1}^1 \frac{x dx}{(1 - \beta x)^4} = \frac{32\pi}{3} \gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})^2
\end{aligned}$$

The total radiation power at a given time t_r for an accelerated charged particle is then

$$\begin{aligned}
P(t_r) &= \frac{q^2}{4\pi c} \left\{ -\frac{4\pi}{3} \gamma^4 [\dot{\beta}^2 + 6\gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}})^2] + 4\pi \gamma^4 \dot{\beta}^2 + \frac{32\pi}{3} \gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})^2 \right\} \\
&= \frac{2q^2}{3c} \gamma^4 [\dot{\beta}^2 + \gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}})^2]
\end{aligned} \tag{29}$$

where γ , $\vec{\beta}$, and $\dot{\vec{\beta}}$ are functions of t_r for the motion of the particle. For $\beta \ll 1$, it is reduced to Larmor formula,

$$P = \frac{2q^2}{3c^3} \dot{v}^2$$

For a particle with charge q and mass m moving in external electric $\vec{E}$ and magnetic $\vec{B}$ fields, the equation of motion of the particle is

$$\begin{cases} \frac{d(\gamma\vec{\beta})}{dt} = \frac{q}{mc} (\vec{E} + \vec{\beta} \times \vec{B}) \\ \frac{d\gamma}{dt} = \frac{q}{mc} \vec{\beta} \cdot \vec{E} \end{cases} \tag{30}$$

and the total radiation power can be obtained from Eqs. (29) and (30) as (HW 14.11),

$$P(t_r) = \frac{2q^4}{3m^2 c^3} \gamma^2 [(\vec{E} + \vec{\beta} \times \vec{B})^2 - (\vec{\beta} \cdot \vec{E})^2] \tag{31}$$

Covariant Formula for $P(t_r)$

To write $P(t_r)$ in Eq. (29) into the covariant form, considering $(p_0, \vec{p}) = \gamma(mc, m\vec{v})$ and $dt = \gamma d\tau$,

$$\frac{d\vec{p}}{dt} = \gamma mc \frac{d\vec{\beta}}{dt} + mc\vec{\beta} \frac{d\gamma}{dt}$$

Thus

$$\frac{d\vec{\beta}}{dt} = \frac{1}{\gamma mc} \left(\frac{d\vec{p}}{d\tau} - mc\vec{\beta} \frac{d\gamma}{d\tau} \right) \frac{d\tau}{dt} = \frac{1}{\gamma^2 mc} \left(\frac{d\vec{p}}{d\tau} - mc\vec{\beta} \frac{d\gamma}{d\tau} \right)$$

and

$$\begin{aligned}
\left[\dot{\beta}^2 + \gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}})^2 \right] &= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{d\vec{p}}{d\tau} \cdot \frac{d\vec{p}}{d\tau} + m^2 c^2 \beta^2 \left(\frac{d\gamma}{d\tau} \right)^2 - 2mc \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{d\gamma}{d\tau} \right] \\
&\quad + \frac{1}{\gamma^2 m^2 c^2} \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} - mc \beta^2 \frac{d\gamma}{d\tau} \right)^2 \\
&= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{d\vec{p}}{d\tau} \cdot \frac{d\vec{p}}{d\tau} + m^2 c^2 \beta^2 \left(\frac{d\gamma}{d\tau} \right)^2 - 2mc \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{d\gamma}{d\tau} \right. \\
&\quad \left. + \gamma^2 \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2 + m^2 c^2 \beta^4 \gamma^2 \left(\frac{d\gamma}{d\tau} \right)^2 - 2mc \beta^2 \gamma^2 \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{d\gamma}{d\tau} \right] \\
&= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{d\vec{p}}{d\tau} \cdot \frac{d\vec{p}}{d\tau} + \beta^2 \gamma^2 \left(\frac{d\gamma mc}{d\tau} \right)^2 - 2\gamma^2 \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{d\gamma mc}{d\tau} + \left(\gamma \vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2 \right] \\
&= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{d\vec{p}}{d\tau} \cdot \frac{d\vec{p}}{d\tau} - \frac{dp_0}{d\tau} \cdot \frac{dp_0}{d\tau} + \gamma^2 \left(\frac{dp_0}{d\tau} \right)^2 - 2\gamma^2 \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{dp_0}{d\tau} + \left(\gamma \vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2 \right] \\
&= \frac{1}{\gamma^4 m^2 c^2} \left[-\frac{dp_\alpha}{d\tau} \frac{dp^\alpha}{d\tau} + \left(\gamma \frac{dp_0}{d\tau} - \gamma \vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2 \right] \\
&= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{1}{m^2 c^2} \left(p_\alpha \frac{dp_\alpha}{d\tau} \right)^2 - \frac{dp_\alpha}{d\tau} \frac{dp^\alpha}{d\tau} \right]
\end{aligned}$$

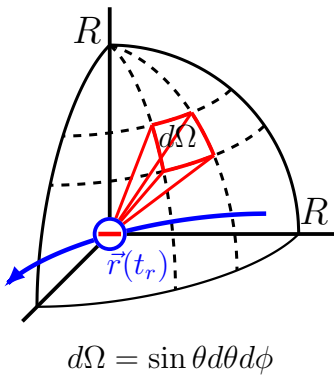
The total radiation power in Eq. (29) can then be written in the covariant form as

$$P(t_r) = \frac{2q^2}{3m^2 c^3} \left[\left(p_\alpha \frac{dp_\alpha}{d\tau} \right)^2 - \frac{dp_\alpha}{d\tau} \frac{dp^\alpha}{d\tau} \right] = \frac{2q^2}{3m^2 c^3} \left[\frac{dp_\alpha}{d\tau} (p^\alpha p_\beta - \delta^\alpha_\beta) \frac{dp^\beta}{d\tau} \right] \quad (32)$$

In Summary

From an accelerated charged particle, the angular distribution of the radiation power and the total radiation power in all directions at a given time t_r of the moving particle are

$$\begin{cases} \frac{dP(t_r)}{d\Omega} = \frac{e^2}{4\pi c} \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{\beta} \cdot \vec{n})^5} \\ P(t_r) = \frac{2q^2}{3c} \gamma^4 [\dot{\beta}^2 + \gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}})^2] \end{cases} \quad (33)$$



where γ , $\vec{\beta}$, and $\dot{\vec{\beta}}$ are functions of t_r for the motion of the particle. If a charged particle moving in an external electromagnetic field $(\vec{E}, \vec{B})$, the radiation power can be written as

$$P(t_r) = \frac{2q^4}{3m^2 c^3} \gamma^2 [(\vec{E} + \vec{\beta} \times \vec{B})^2 - (\vec{\beta} \cdot \vec{E})^2] \quad (34)$$

In the following, we will discuss a few special cases of radiation.

14.4. Examples of the Radiation by a Charged Particle in Acceleration

In this section, we will study the radiation of a charged particle when its acceleration is in parallel with, perpendicular to, or in an arbitrary angle with respect to its velocity.

(a) Non-relativistic case $\beta \ll 1$

Since $\vec{n} - \vec{\beta} \simeq \vec{n}$ and $(1 - \vec{\beta} \cdot \vec{n}) \simeq 1$, from Eq. (33), the angular distribution of the radiation power is

$$\frac{dP}{d\Omega} \simeq \frac{e^2}{4\pi c} \left| \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \right|^2 = \frac{e^2}{4\pi c} \left[\dot{\vec{\beta}} \cdot \dot{\vec{\beta}} - (\vec{n} \cdot \dot{\vec{\beta}})^2 \right] = \frac{e^2 \dot{v}^2}{4\pi c^3} \sin^2 \theta \quad (35)$$

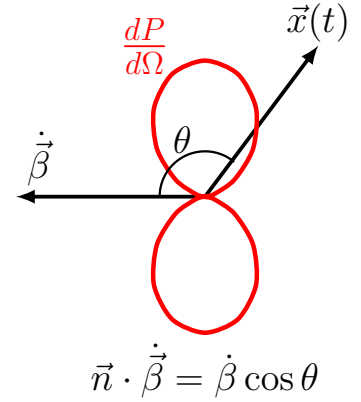
where

$$\begin{aligned} \left(\frac{dP}{d\Omega} \right)_{max} &= \frac{e^2 \dot{v}^2}{4\pi c^3} & \text{at } \theta = \frac{\pi}{2} \\ \left(\frac{dP}{d\Omega} \right)_{min} &= 0 & \text{at } \theta = 0, \pi \end{aligned}$$

The total radiation power is

$$P = \int \frac{dP}{d\Omega} d\Omega = \frac{e^2 \dot{v}^2}{4\pi c^3} \int \sin^3 \theta d\theta d\phi = \frac{2e^2}{3c^3} \dot{v}^2 \quad (36)$$

This is the Larmor formula.



(b) $\vec{\beta} \parallel \dot{\vec{\beta}}$ — Charged-Particle Beams in Linear Accelerators

With $\vec{\beta} \times \dot{\vec{\beta}} = 0$,

$$\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}] = \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) = (\vec{n} \cdot \dot{\vec{\beta}}) \vec{n} - \dot{\vec{\beta}}$$

and

$$\begin{aligned} |\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2 &= |(\vec{n} \cdot \dot{\vec{\beta}}) \vec{n} - \dot{\vec{\beta}}|^2 \\ &= \dot{\beta}^2 - (\vec{n} \cdot \dot{\vec{\beta}})^2 = \dot{\beta}^2 \sin^2 \theta \end{aligned}$$

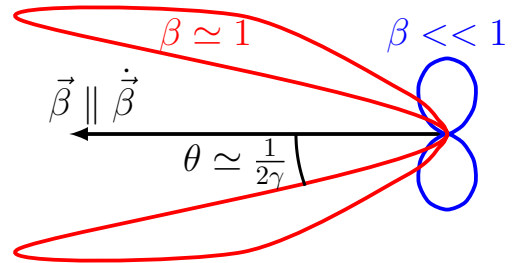
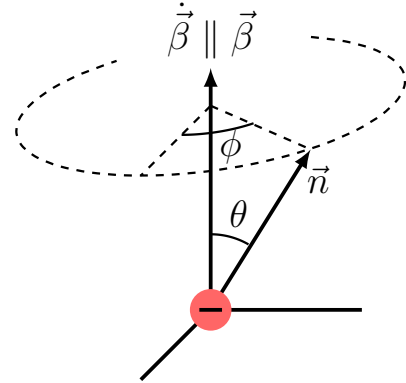
the angular distribution of the radiation power is

$$\frac{dP(t_r)}{d\Omega} = \frac{e^2 \dot{v}^2}{4\pi c^3} \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5} \quad (37)$$

The radiation is therefore rotationally symmetric around $\dot{\vec{\beta}} \parallel \vec{\beta}$ (independent of ϕ). For $\beta \ll 1$, Eq. (37) reduces to the Larmor formula in Eq. (36).

The minimal radiation is at $\theta = 0$ where

$$\left(\frac{dP(t_r)}{d\Omega} \right)_{min, \theta=0} = 0$$



The maximal radiation occurs at θ_{max} where θ_{max} is the solution of

$$0 = \frac{d}{d\theta} \left[\frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5} \right] = \frac{\sin \theta}{(1 - \beta \cos \theta)^6} [3\beta \cos^2 \theta + 2 \cos \theta - 5\beta]$$

which yields

$$\cos \theta_{max} = \frac{1}{3\beta} \left(\sqrt{1 + 15\beta^2} - 1 \right)$$

For $\beta \ll 1$, $\theta_{max} = 90^\circ$ and

$$\left(\frac{dP(t_r)}{d\Omega} \right)_{max, \beta \rightarrow 0} = \frac{e^2 \dot{v}^2}{4\pi c^3}$$

For $\beta \simeq 1$, we can expand $\cos \theta_{max}$ with a small parameter

$$0 < \epsilon = (1 - \beta) \ll 1$$

Since

$$\cos \theta_{max} = \frac{1}{3(1 - \epsilon)} \left[\sqrt{1 + 15(1 - \epsilon)^2} - 1 \right] \simeq 1 - \frac{\epsilon}{4} \sim 1 \quad \Rightarrow \quad 0 < \theta_{max} \ll 1$$

for $\beta \simeq 1$, θ_{max} can be solved from

$$1 - \frac{\theta_{max}^2}{2} \simeq \cos \theta_{max} \simeq 1 - \frac{\epsilon}{4} \quad \Rightarrow \quad \theta_{max}^2 \simeq \frac{\epsilon}{2}$$

For $\beta \simeq 1$ and $\gamma \gg 1$, therefore,

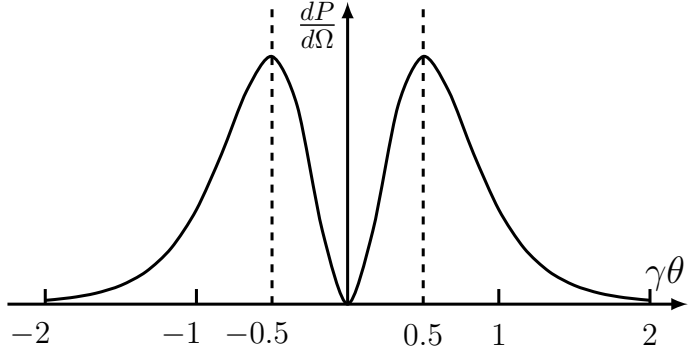
$$\theta_{max}^2 \simeq \frac{1 - \beta}{2} = \frac{(1 - \beta)(1 + \beta)}{2(1 + \beta)} \simeq \frac{1}{4\gamma^2} \quad \Rightarrow \quad \theta_{max} \simeq \pm \frac{1}{2\gamma} \ll 1$$

and the radiation is concentrated inside a small cone of $\theta \ll 1$. From Eq. (37), the angular distribution of the radiation power at $\gamma \gg 1$ can be simplified as

$$\begin{aligned} \frac{dP(t_r)}{d\Omega} &= \frac{e^2 \dot{v}^2}{4\pi c^3} \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5} \\ &\stackrel{\theta \ll 1}{\simeq} \frac{e^2 \dot{v}^2}{4\pi c^3} \frac{\theta^2}{[1 - (1 - 1/2\gamma^2)(1 - \theta^2/2)]^5} = \frac{e^2 \dot{v}^2}{4\pi c^3} \frac{(4\gamma^2)^5 \theta^2}{[2 + (2\gamma^2 - 1)\theta^2]^5} \\ &\stackrel{\gamma^2 \gg 1}{\simeq} \frac{8e^2 \dot{v}^2}{\pi c^3} \frac{\gamma^8 (\gamma\theta)^2}{[1 + (\gamma\theta)^2]^5} \end{aligned} \quad (38)$$

where $\beta \simeq 1 - 1/(2\gamma^2)$, and

$$\left(\frac{dP(t_r)}{d\Omega} \right)_{max, \beta \rightarrow 1} \stackrel{\gamma\theta=1/2}{=} \frac{8e^2 \dot{v}^2}{\pi c^3} \frac{\gamma^8 (1/2)^2}{[1 + (1/2)^2]^5} = \frac{2^{14}}{5^5} \left(\frac{e^2 \dot{v}^2}{4\pi c^3} \right) \gamma^8 \quad (39)$$



Total Radiation Power in all the Directions for the Case of $\vec{\beta} \parallel \dot{\vec{\beta}}$

$$\begin{aligned}
P(t_r) &= \int \frac{dP(t_r)}{d\Omega} d\Omega = \frac{e^2 \dot{v}^2}{2c^3} \int_0^\pi \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5} \sin \theta d\theta = \frac{e^2 \dot{v}^2}{2c^3} \int_{-1}^1 \frac{1 - x^2}{(1 - \beta x)^5} dx \\
&\stackrel{y=1-\beta x}{=} \frac{e^2 \dot{v}^2}{2c^3 \beta^3} \int_{1-\beta}^{1+\beta} \frac{\beta^2 - (1-y)^2}{y^5} dy = \frac{e^2 \dot{v}^2}{2c^3 \beta^3} \int_{1-\beta}^{1+\beta} \left(\frac{\beta^2 - 1}{y^5} + \frac{2}{y^4} - \frac{1}{y^3} \right) dy \\
&= \frac{2e^2}{3c^3} \gamma^6 \dot{v}^2 = \frac{2e^2}{3c^3 m^2} \left| \frac{d\vec{p}}{dt_r} \right|^2 \tag{40}
\end{aligned}$$

where

$$\begin{aligned}
\frac{d\vec{p}}{dt_r} &= \frac{d\gamma m \vec{v}}{dt_r} = \gamma m \dot{\vec{v}} + m \vec{v} \dot{\gamma} = \gamma m c \dot{\vec{\beta}} + m \vec{v} \frac{\dot{\vec{\beta}} \cdot \vec{\beta}}{(1 - \beta^2)^{3/2}} \\
&= \gamma m c \left[\dot{\vec{\beta}} + \gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}}) \vec{\beta} \right] \stackrel{\vec{\beta} \parallel \dot{\vec{\beta}}}{=} \gamma m \dot{\vec{v}} (1 + \gamma^2 \beta^2) = \gamma^3 m \dot{\vec{v}}
\end{aligned}$$

Since for the energy of a charged particle, $E = \gamma m c^2$ and $E^2 = m^2 c^4 + c^2 \vec{p} \cdot \vec{p}$,

$$c^2 \vec{p} \cdot \frac{d\vec{p}}{dt_r} = E \frac{dE}{dt_r} = E \frac{dE}{d\vec{r}} \cdot \frac{d\vec{r}}{dt_r} = \gamma m c^2 \vec{v} \cdot \frac{d\vec{r}}{d\vec{r}} = c^2 \vec{p} \cdot \frac{d\vec{r}}{d\vec{r}}$$

Therefore,

$$\vec{p} \cdot \frac{d\vec{p}}{dt_r} = \vec{p} \cdot \frac{d\vec{r}}{d\vec{r}} \implies \frac{d\vec{p}}{dt_r} = \frac{d\vec{r}}{d\vec{r}} \quad \text{for } \vec{r} \parallel \vec{p}$$

For $\vec{p} \parallel \dot{\vec{p}} \parallel \vec{r}$, the particle moves in one-dimensional and we can choose $\vec{r} = (x, 0, 0)$. Then the total radiation power during the acceleration of a charged particle in one-dimensional motion is

$$P(t_r) = \frac{2e^2}{3c^3 m^2} \left(\frac{dE}{dx} \right)^2 \tag{41}$$

where dE/dx is the rate of the energy gain or loss due to the acceleration during the motion.

Energy Gain/Loss in Linear Accelerators

In linear accelerators (linac), the radiation occurs during acceleration of electrons and $dE/dx = dE_{\text{gain}}/dx$ in Eq. (41) is the energy gain per meter for an electron during the acceleration, *i.e.*

$$P(t_r) = \frac{2e^2}{3c^3 m^2} \left(\frac{dE_{\text{gain}}}{dx} \right)^2$$

From energy conservation, this radiation power equals the energy loss per second during the electron acceleration, *i.e.*

$$P(t_r) = \frac{dE_{\text{loss}}}{dt_r} = \frac{dE_{\text{loss}}}{dx} \frac{dx}{dt_r} = \frac{dE_{\text{loss}}}{dx} \beta c$$

Therefore

$$\frac{dE_{loss}}{dx} = \frac{2}{3\beta} \left(\frac{e}{mc^2} \right)^2 \left(\frac{dE_{gain}}{dx} \right)^2$$

and

$$\frac{\text{energy loss}}{\text{energy gain}} = \frac{dE_{loss}/dx}{dE_{gain}/dx} = \frac{2}{3\beta} \left(\frac{e}{mc^2} \right)^2 \left(\frac{dE_{gain}}{dx} \right)$$

where for electron, $mc^2 \simeq 0.5 \text{ MeV}$ and $(e^2/mc^2) \simeq 3 \times 10^{-15} \text{ m}$.

For ILC, $\frac{dE_{gain}}{dx} \sim 50 \text{ MeV/m}$

$$\frac{\text{energy loss}}{\text{energy gain}} = \frac{2}{3\beta} \frac{3 \times 10^{-15} \text{ m} \times 50 \text{ MeV/m}}{0.5 \text{ MeV}} \sim \frac{10^{-13}}{\beta} \ll 10^{-5}$$

The energy loss of an electron beam due to the radiation during the acceleration is therefore insignificant in linear accelerators.

(c) A Charged Particle Moving in Instantaneously Circular Motion — Synchrotron Radiation of Particle Beam in a Circular Accelerator

The radiation power per unit solid angle from an accelerated charged particle is

$$\frac{dP(t_r)}{d\Omega} = \frac{e^2}{4\pi c} \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{\beta} \cdot \vec{n})^5}$$

For circular motion, $\vec{\beta} \perp \dot{\vec{\beta}}$, we can choose a coordinate such that $\vec{\beta}$ is aligned with $\vec{e}_3$ and $\dot{\vec{\beta}}$ is aligned with $\vec{e}_2$. The circular orbit is in the $\vec{e}_2$ - $\vec{e}_3$ plane and the center of the circle is on the $\vec{e}_2$ axis. Since

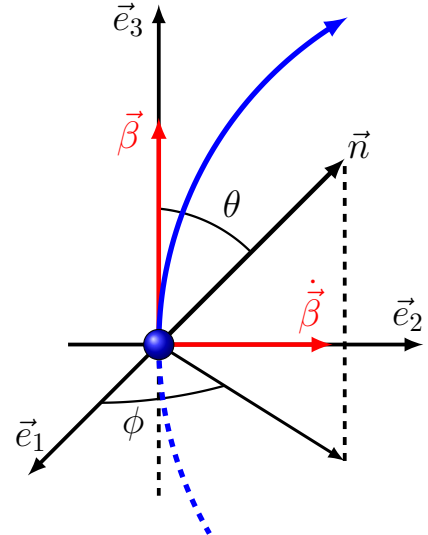
$$\vec{n} \cdot \vec{\beta} = \beta \cos \theta \quad \text{and} \quad \vec{n} \cdot \dot{\vec{\beta}} = \dot{\beta} \sin \theta \sin \phi$$

$$\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}] = (\vec{n} \cdot \dot{\vec{\beta}})(\vec{n} - \vec{\beta}) - (1 - \vec{n} \cdot \vec{\beta})\dot{\vec{\beta}}$$

$$\begin{aligned} |\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2 &= \dot{\beta}^2 (1 - \vec{n} \cdot \vec{\beta})^2 - (\vec{n} \cdot \dot{\vec{\beta}})^2 (1 - \beta^2) \\ &= \dot{\beta}^2 [(1 - \beta \cos \theta)^2 - (1 - \beta^2) \sin^2 \theta \sin^2 \phi] \end{aligned}$$

Therefore, the angular distribution of the radiation from a particle in circular motion is

$$\frac{dP(t_r)}{d\Omega} = \frac{e^2}{4\pi c} \frac{\dot{\beta}^2}{(1 - \beta \cos \theta)^3} \left[1 - \frac{\sin^2 \theta \sin^2 \phi}{\gamma^2 (1 - \beta \cos \theta)^2} \right] \quad (42)$$



Maximal and Minimal Radiation

The maximal radiation is at θ_{max} that is the solution of

$$\begin{aligned} 0 &= \frac{d}{d\theta} \left[\frac{1}{(1 - \beta \cos \theta)^3} - \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5} \left(\frac{\sin^2 \phi}{\gamma^2} \right) \right] \\ &= \frac{-\sin \theta}{(1 - \beta \cos \theta)^4} \left\{ 3\beta - \left[\frac{2 \cos \theta}{(1 - \beta \cos \theta)} - \frac{5\beta \sin^2 \theta}{(1 - \beta \cos \theta)^2} \right] \left(\frac{\sin^2 \phi}{\gamma^2} \right) \right\} \end{aligned} \quad (43)$$

For arbitrary ϕ , the roots are

$$\sin \theta = 0 \quad \implies \quad \theta_{max} = 0 \quad \text{and} \quad \theta_{min} = \pi$$

where for $\gamma \gg 1$, the 2nd term in Eq. (43) is much smaller than 3β of the 1st term. From Eq. (42) with $\sin \theta = 0$, the maximal and minimal radiation power are

$$\begin{cases} \left(\frac{dP(t_r)}{d\Omega} \right)_{max} = \frac{e^2}{4\pi c} \frac{\dot{\beta}^2}{(1 - \beta)^3} & \text{at } \theta = \theta_{max} = 0 \\ \left(\frac{dP(t_r)}{d\Omega} \right)_{min} = \frac{e^2}{4\pi c} \frac{\dot{\beta}^2}{(1 + \beta)^3} & \text{at } \theta = \theta_{min} = \pi \end{cases}$$

where for the uniform circular motion, $v = \text{constant}$ and $\gamma = \text{constant}$,

$$m\gamma \frac{d\vec{v}}{dt_t} = \frac{d}{dt_t}(m\gamma\vec{v}) = \frac{d\vec{p}}{dt_r}$$

For ultra-relativistic case, $\gamma \gg 1$,

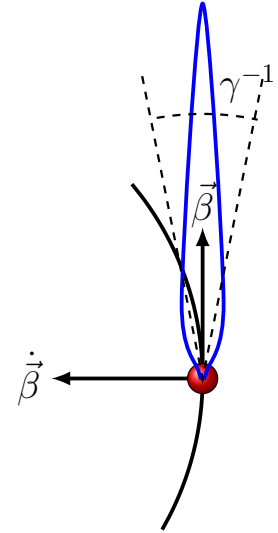
$$\beta = \left(1 - \frac{1}{\gamma^2}\right)^{1/2} \simeq 1 - \frac{1}{2\gamma^2} \quad \implies \quad \begin{cases} (1 - \beta) = 1/(2\gamma^2) \\ (1 + \beta) = 2 \end{cases}$$

and

$$\begin{cases} \left(\frac{dP(t_r)}{d\Omega} \right)_{max} \stackrel{\gamma \gg 1}{=} \frac{2e^2}{\pi c^3} \gamma^6 \dot{v}^2 \\ \left(\frac{dP(t_r)}{d\Omega} \right)_{min} \stackrel{\gamma \gg 1}{=} \frac{e^2}{4\pi c^3} \frac{\dot{v}^2}{2^3} \end{cases}$$

Therefore,

$$\left(\frac{dP(t_r)}{d\Omega} \right)_{max} / \left(\frac{dP(t_r)}{d\Omega} \right)_{min} \sim \gamma^6$$



Radiation Cone for $\gamma \gg 1$

For $\gamma \gg 1$ and $\theta \ll 1$, $\beta \simeq 1 - 1/(2\gamma^2)$, $\cos \theta \simeq 1 - \theta^2/2$, and for $dP/d\Omega$ in Eq. (42)

$$\begin{cases} (1 - \beta \cos \theta) \simeq 1 - (1 - 1/(2\gamma^2))(1 - \theta^2/2) \simeq \frac{1}{2\gamma^2}[1 + (\gamma\theta)^2] \\ \sin \theta \simeq 0 \end{cases}$$

The angular distribution of the radiation from a charged particle in circular motion with $\gamma \gg 1$ is then

$$\frac{dP(t_r)}{d\Omega} = \frac{e^2}{4\pi c} \frac{\dot{\beta}^2}{(1 - \beta \cos \theta)^3} \left[1 - \frac{\sin^2 \theta \sin^2 \phi}{\gamma^2 (1 - \beta \cos \theta)^2} \right] \simeq \frac{2e^2}{\pi c} \frac{\gamma^6 \dot{\beta}^2}{[1 + (\gamma\theta)^2]^3} \quad (44)$$

where the radiation decreases from its maximum at $\theta = 0$ quickly when $\gamma\theta > 1$. In this case, the radiation is again concentrated in a forward cone of $\Delta\theta \sim 1/\gamma$.

Total Radiation Power of a Charged Particle in Circular Motion

From the angular distribution of the radiation in Eq. (42), the total radiation power can be calculated from the integration

$$P(t_r) = \int \frac{dP}{d\Omega} d\Omega = \frac{e^2 \dot{v}^2}{4\pi c^3} \int \left[\frac{1}{(1 - \beta \cos \theta)^3} - \frac{\sin^2 \phi \sin^2 \theta}{\gamma^2 (1 - \beta \cos \theta)^5} \right] \sin \theta d\theta d\phi$$

where

$$\int_0^\pi \frac{1}{(1 - \beta \cos \theta)^3} \sin \theta d\theta = 2\gamma^4 \quad \text{and} \quad \int_0^\pi \frac{\sin^2 \phi \sin^2 \theta}{\gamma^2 (1 - \beta \cos \theta)^5} \sin \theta d\theta = \frac{4}{3}\gamma^4$$

In a circular orbit, the total radiation power from a charged particle is thus

$$P(t_r) = \frac{2e^2}{3c^3} \gamma^4 \dot{v}^2 = \frac{2e^2}{3m^2 c^3} \gamma^2 \left| \frac{d\vec{p}}{dt_r} \right|^2 \quad (45)$$

In comparison, the total radiation power from a charge in linear motion [in Eq. (40)] is

$$P(t_r) = \frac{2e^2}{3m^2 c^3} \left| \frac{d\vec{p}}{dt_r} \right|^2$$

The energy loss from beam particles in a circular accelerator is thus γ^2 more than that in a linear accelerator if the two have a comparable acceleration force.

For a circular orbit, $\dot{v} = v^2/\rho$ where ρ is the radius of curvature of the orbit, the total radiation power from a charged particle can be written as

$$P(t_r) = \frac{2e^2 c}{3\rho^2} \gamma^4 \beta^4 \quad (46)$$

The radiation is therefore inversely proportional to the square of the radius of the trajectory and proportional to the 4th power of the kinetic energy of the particle.

Energy Loss due to Radiation in Circular Accelerators

In circular accelerators, $\dot{v} = v^2/\rho$, where $2\pi\rho$ is the circumference of a circular accelerator and Eq. (45) becomes

$$\begin{aligned} P(t_r) &= \frac{2e^2 c}{3\rho^2} \gamma^4 \beta^4 \quad \gamma \gg 1 \quad \frac{8\pi^2 e^2 c}{3} \frac{\gamma^4}{(2\pi\rho)^2} \\ &= \left[\frac{8\pi^2 c}{3} \left(\frac{e^2}{m_e c^2} \right) (m_e c^2) \right] \frac{\gamma^4}{(2\pi\rho)^2} = (11 \text{ m}^2 \cdot \text{eV/s}) \times \frac{\gamma^4}{(2\pi\rho)^2} \end{aligned}$$

where $r_e = [e^2/m_e c^2]_{\text{Gaussian-unit}} = [(1/2\pi\epsilon_0)(e^2/m_e c^2)]_{\text{SI-unit}} = 2.8 \times 10^{-15} \text{ m}$ is the electron classical radius and $m_e c^2 \simeq 0.5 \text{ MeV}$ is the electron rest mass.

The energy loss per revolution of a charged-particle beam in a circular motion due to the radiation is then

$$E_{\text{loss}}/\text{revolution} = P(t_r) \times \frac{2\pi\rho}{c} = \frac{8\pi^2 e^2}{3} \frac{\gamma^4}{2\pi\rho} = (3.7 \times 10^{-8} \text{ m}\cdot\text{eV}) \times \frac{\gamma^4}{2\pi\rho}$$

machine	E	mc^2	$\gamma = E/mc^2$	$2\pi\rho$	$P(t_r)$	$E_{\text{loss}}/\text{revolution}$
LEP (e)	100 GeV	0.5 MeV	2×10^5	27 km	$3 \times 10^4 \text{ GeV/s}$	$2 \text{ GeV} \simeq 2\%E$
LHC (p)	7000 GeV	940 MeV	7×10^3	27 km	40 MeV/s	4 KeV

Question: can we build a “LHC” with electron beam?

For 500GeV electron beam in LHC ring,

$$\gamma \simeq 10^6 \quad \implies \quad E_{\text{loss}}/\text{revolution} \sim 1.4 \times 10^3 \text{ GeV} \sim 300\%E$$

(d) Acceleration in Arbitrary Direction

In general, the acceleration of a particle motion can be written as an addition of two accelerations that are parallel and perpendicular to the velocity, respectively, *i.e.*

$$\dot{\vec{\beta}} = \dot{\vec{\beta}}_{\parallel} + \dot{\vec{\beta}}_{\perp} \quad (47)$$

where $\dot{\vec{\beta}}_{\parallel} \parallel \vec{\beta}$ and $\dot{\vec{\beta}}_{\perp} \perp \vec{\beta}$. The power radiated per solid angle at the retarded time t_r is

$$\frac{dP(t_r)}{d\Omega} = \frac{e^2}{4\pi c} \left\{ \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{n} \cdot \vec{\beta})^5} \right\}$$

Since

$$\begin{aligned} \left| \vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}] \right|^2 &= \left| \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}_{\parallel}) \right|^2 + \left| \vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}_{\perp}] \right|^2 \\ &\quad + 2 \left[\vec{n} \times (\vec{n} \times \dot{\vec{\beta}}_{\parallel}) \right] \cdot \left\{ \vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}_{\perp}] \right\} \end{aligned}$$

and the 3rd term, the cross term, can be written as

$$\begin{aligned} &\left[\vec{n} \times (\vec{n} \times \dot{\vec{\beta}}_{\parallel}) \right] \cdot \left\{ \vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}_{\perp}] \right\} \\ &= \left[\vec{n}(\vec{n} \cdot \dot{\vec{\beta}}_{\parallel}) - \dot{\vec{\beta}}_{\parallel} \right] \cdot \left[(\vec{n} - \vec{\beta})(\vec{n} \cdot \dot{\vec{\beta}}_{\perp}) - \dot{\vec{\beta}}_{\perp}(1 - \vec{n} \cdot \vec{\beta}) \right] \\ &= (\vec{\beta} \cdot \dot{\vec{\beta}} - \vec{n} \cdot \dot{\vec{\beta}}_{\parallel})(\vec{n} \cdot \dot{\vec{\beta}}_{\perp}) \end{aligned}$$

$$\frac{dP(t_r)}{d\Omega} = \left[\frac{dP(t_r)}{d\Omega} \right]_{\parallel} + \left[\frac{dP(t_r)}{d\Omega} \right]_{\perp} + \frac{e^2}{2\pi c} \frac{(\vec{\beta} \cdot \dot{\vec{\beta}} - \vec{n} \cdot \dot{\vec{\beta}}_{\parallel})(\vec{n} \cdot \dot{\vec{\beta}}_{\perp})}{(1 - \vec{n} \cdot \vec{\beta})^5} \quad (48)$$

where the first two terms are the radiation power in solid angle in the cases of $\dot{\vec{\beta}} = \dot{\vec{\beta}}_{\parallel}$ and $\dot{\vec{\beta}} = \dot{\vec{\beta}}_{\perp}$, respectively, and have already been calculated in the previous two sections. The third term in Eq. (48) is the interference term between the radiations of $\dot{\vec{\beta}}_{\parallel}$ and $\dot{\vec{\beta}}_{\perp}$. We now calculate the contribution of the third term. With a coordinate where $\vec{\beta} \perp \vec{e}_3$,

$$\vec{\beta} = \beta \vec{e}_3, \quad \vec{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta), \quad \dot{\vec{\beta}} = (\dot{\beta}_1, \dot{\beta}_2, \dot{\beta}_3),$$

and

$$(\vec{\beta} \cdot \dot{\vec{\beta}} - \vec{n} \cdot \dot{\vec{\beta}}_{\parallel})(\vec{n} \cdot \dot{\vec{\beta}}_{\perp}) = \dot{\beta}_3(\beta - \cos \theta)(\dot{\beta}_1 \cos \phi + \dot{\beta}_2 \sin \phi) \sin \theta$$

The interference term in Eq. (48) is then

$$\left[\frac{dP(t')}{d\Omega} \right]_{\parallel \times \perp} = \frac{e^2}{2\pi c} \frac{\dot{\beta}_3(\beta - \cos \theta) \sin \theta}{(1 - \beta \cos \theta)^5} (\dot{\beta}_1 \cos \phi + \dot{\beta}_2 \sin \phi)$$

and its contribution to the total radiation power is

$$\int \left[\frac{dP(t')}{d\Omega} \right]_{\parallel \times \perp} d\Omega \sim \int_0^{2\pi} (\dot{\beta}_1 \cos \phi + \dot{\beta}_2 \sin \phi) d\phi = 0$$

The total radiation power for a charged particle accelerating in an arbitrary direction is therefore

$$P(t') = \int \left[\frac{dP(t')}{d\Omega} \right]_{\parallel} d\Omega + \int \left[\frac{dP(t')}{d\Omega} \right]_{\perp} d\Omega = P_{\parallel}(t') + P_{\perp}(t')$$

where

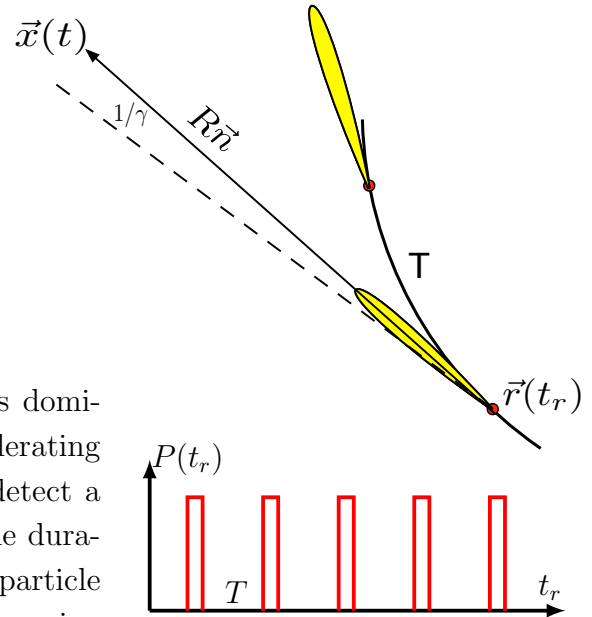
$$P_{\parallel}(t') = \frac{2e^2}{3m^2c^3} \left| \frac{d\vec{p}}{dt} \right|^2 \quad \text{for } \dot{\vec{\beta}} = \dot{\vec{\beta}}_{\parallel}$$

$$P_{\perp}(t') = \frac{2e^2}{3m^2c^3} \left| \frac{d\vec{p}}{dt} \right|^2 \gamma^2 \quad \text{for } \dot{\vec{\beta}} = \dot{\vec{\beta}}_{\perp}$$

For ultra-relativistic particles,

$$P_{\perp}(t')/P_{\parallel}(t') = \gamma^2 \gg 1$$

Therefore, the perpendicular component of $\dot{\vec{\beta}}$ is dominant the radiation. For a relativistic particle accelerating in arbitrary motion, a stationary observer will detect a pulse or burst of radiation with a very short time duration due to a narrow $(1/\gamma)$ radiation cone. If a particle is in a periodic motion, the observer will see a succession of short radiation bursts. During each radiation burst seen by an observer, the particle travels a very short distance because of the very narrow radiation cone and, therefore, the directional vector $\vec{n}$ from the position of particle to the observer is approximately constant.



Characteristics of the radiation field emitted by a relativistic charged particle:

- (a) The radiation due to $\dot{\vec{\beta}}_{\perp}$ is dominant.
- (b) The radiation is concentrated in the forward direction along $\vec{\beta}$ in a narrow cone of $1/\gamma$ opening.
- (c) For a stationary observer, the observed radiation of a moving charge is a short pulse. The radiation is only visible when the particle's velocity is directed toward the observer. During the observed pulse, the position of the particle $\vec{r}(t_r)$ is almost constant and $\vec{n}$ is approximately a constant vector.

14.5 Frequency Spectrum of Radiation Energy from a Relativistic Charge

To obtain the frequency spectrum of the radiation from an accelerated charged particle, we need to calculate the radiation energy per unit frequency, which is the energy density in the frequency space, as a function of the radiation frequency. For that purpose, the Fourier transformation of the time dependence of the radiation field needs to be calculated at a fixed observation point, not at a given particle position. We need therefore go back to the original formula for the radiation power per unit solid angle observed at observation point $\vec{x}$ and time t in Eq. (22) with the retardation condition,

$$\frac{dP(\vec{x}, t)}{d\Omega} = \frac{c}{4\pi} R^2 |\vec{E}|^2 = \frac{e^2}{4\pi c} \left| \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{\beta} \cdot \vec{n})^3} \right|_{ret}^2 \quad (49)$$

where and the retarded time is calculated at the particle position $\vec{r}(t_r)$ with $t_r = t - |\vec{x} - \vec{r}(t_r)|/c$. For convenience, we use the notation

$$\vec{A}(t) = \frac{e}{\sqrt{4\pi c}} \left\{ \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{\beta} \cdot \vec{n})^3} \right\}_{ret} \quad \text{and} \quad \frac{dP(t)}{d\Omega} = |\vec{A}(t)|^2 \quad (50)$$

For an ultra-relativistic charged particle moving in an arbitrary trajectory, the radiation energy due to $\dot{\vec{\beta}}_{\parallel}$ is negligible as compare to the radiation energy due to $\dot{\vec{\beta}}_{\perp}$. We need thus only consider the radiation due to $\dot{\vec{\beta}}_{\perp}$ which is equivalent to consider a particle moving in an instantaneous circular motion. In the following, we will calculate the Fourier transformation of $\vec{A}(t)$,

$$\vec{A}(\omega) = \int_{-\infty}^{\infty} \vec{A}(t) e^{i\omega t} dt \quad \text{and} \quad \vec{A}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \vec{A}(\omega) e^{-i\omega t} d\omega$$

Note that for an observer at a fixed location $\vec{x}$, the radiation can only be observed in a very short duration of the radiation burst when the observer is inside a radiation cone of the particle. Therefore, $\vec{A}(t)$ is nonzero only for a very small interval of t in the Fourier integral

for $\vec{A}(\omega)$. Let $dE(\vec{x}, t)/d\Omega$ be the radiation energy per unit solid angle observed at $(\vec{x}, t)$, *i.e.*

$$\frac{dP(t)}{d\Omega} = \frac{d^2E}{dt d\Omega} \quad (51)$$

Then

$$\begin{aligned} \frac{dE}{d\Omega} &= \int \frac{dP(t)}{d\Omega} dt = \int |\vec{A}(t)|^2 dt \\ &= \frac{1}{4\pi^2} \int dt \int_{-\infty}^{\infty} e^{i\omega_1 t} d\omega_1 \int_{-\infty}^{\infty} e^{-i\omega_2 t} d\omega_2 [\vec{A}^*(\omega_1) \cdot \vec{A}(\omega_2)] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |\vec{A}(\omega)|^2 d\omega = \frac{1}{\pi} \int_0^{\infty} |\vec{A}(\omega)|^2 d\omega \end{aligned}$$

where $\vec{A}^*(\omega) = \vec{A}(-\omega)$ for real $\vec{A}(t)$. The radiation energy per unit solid angle per unit frequency is thus connected to the Fourier transformation of $\vec{A}(t)$ as

$$\frac{d^2E}{d\omega d\Omega} = \frac{1}{\pi} |\vec{A}(\omega)|^2 \quad (52)$$

The frequency spectrum of the radiation from a charged particle can therefore be obtained with the Fourier transformation of $\vec{A}(t)$. Note that $|\vec{A}(\omega)|^2$ is not directly related to the Fourier transformation of $dP(t)/d\Omega$, since

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{dP(t)}{d\Omega} e^{i\omega t} dt &= \int_{-\infty}^{\infty} e^{i\omega t} dt \int_{-\infty}^{\infty} e^{i\omega_1 t} d\omega_1 \int_{-\infty}^{\infty} e^{-i\omega_2 t} d\omega_2 [\vec{A}^*(\omega_1) \cdot \vec{A}(\omega_2)] \\ &= 2\pi \int_{-\infty}^{\infty} d\omega_1 \int_{-\infty}^{\infty} d\omega_2 [\vec{A}^*(\omega_1) \cdot \vec{A}(\omega_2)] \delta(\omega + \omega_1 - \omega_2) \\ &= 2\pi \int_{-\infty}^{\infty} d\omega_1 [\vec{A}^*(\omega_1) \cdot \vec{A}(\omega + \omega_1)] \end{aligned}$$

We now calculate the Fourier transformation of $\vec{A}(t)$,

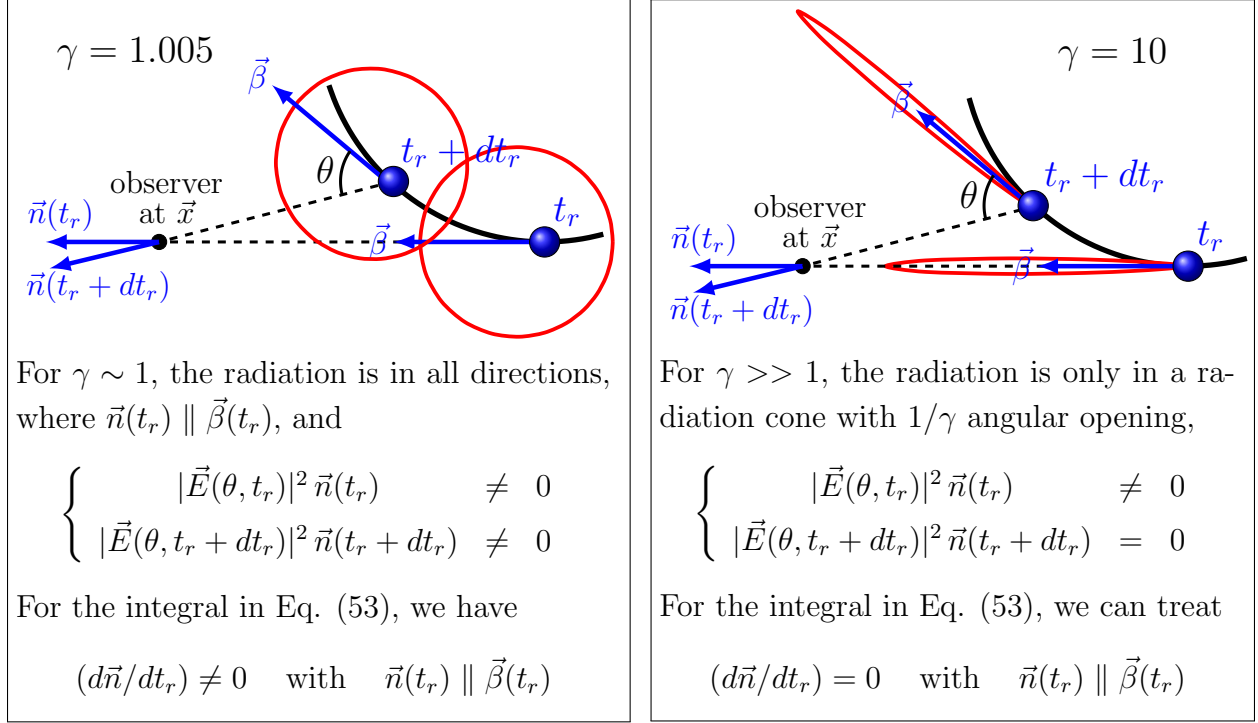
$$\vec{A}(\omega) = \frac{e}{\sqrt{4\pi c}} \int_{-\infty}^{\infty} \left\{ \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{\beta} \cdot \vec{n})^3} \right\}_{ret} e^{i\omega t} dt \quad (53)$$

where the integral is over the observer's time t , while $\vec{\beta}$, $\dot{\vec{\beta}}$, and $\vec{n}$ are functions of the retarded time t_r . It is convenient to change the integration variable from t to t_r , where

$$t = t_r + \frac{1}{c} |\vec{x} - \vec{r}(t_r)| = t_r + \frac{1}{c} [\vec{x} - \vec{r}(t_r)] \cdot \vec{n} \quad (54)$$

and

$$\frac{dt}{dt_r} = (1 - \vec{\beta} \cdot \vec{n}) + \frac{1}{c} [\vec{x} - \vec{r}(t_r)] \cdot \frac{d\vec{n}}{dt_r} \quad (55)$$



Note that the integrand in Eq. (53), which is proportional to the radiation field $\vec{E}(\vec{x}, t)$ at the observation point, is nonzero only when $\vec{n}$ is inside a radiation cone of the particle. For an ultra-relativistic particle, the angular width of the radiation cone is $\sim 1/\gamma$ with $\gamma \gg 1$. Due to such a narrow opening, a radiation cone can only be aligned with $\vec{n}$ for a very short period of t when the particle moves ultra-relativistically. To the observer at $\vec{x}$, this very short period of the alignment of the radiation cone to the observer produces a very short radiation pulse. During this very short radiation pulse, the direction change of $\vec{n}$ due to the motion of the particle is negligible, *i.e.* for the integral in Eq. (53)

$$\frac{d\vec{n}}{dt} \simeq 0 \quad \text{and} \quad \frac{d\vec{n}}{dt_r} = \left(\frac{d\vec{n}}{dt} \right) \left(\frac{dt}{dt_r} \right) \simeq 0$$

For an ultra-relativistic particle, the change from dt to dt_r in Eq. (55) can therefore be simplified as

$$dt \simeq (1 - \vec{\beta} \cdot \vec{n}) dt_r \quad (56)$$

and, with t_r as the integration variable, the Fourier transformation of $\vec{A}(t)$ in Eq. (53) can be written as

$$\vec{A}(\omega) \simeq \frac{e}{\sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} \int_{-\infty}^{\infty} \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{\beta} \cdot \vec{n})^2} e^{i\omega[t_r - \vec{n} \cdot \vec{r}(t_r)/c]} dt_r \quad (57)$$

With $d\vec{n}/dt_r \simeq 0$,

$$\frac{d}{dt_r} \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{1 - \vec{\beta} \cdot \vec{n}} \right] = \frac{\vec{n} \times (\vec{n} \times \dot{\vec{\beta}})}{1 - \vec{\beta} \cdot \vec{n}} + \frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{(1 - \vec{\beta} \cdot \vec{n})^2} (\dot{\vec{\beta}} \cdot \vec{n}) = \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{\beta} \cdot \vec{n})^2}$$

and, therefore, the Fourier transformation of $\vec{A}(t)$ can be written as

$$\begin{aligned}\vec{A}(\omega) &\simeq \frac{e}{\sqrt{4\pi c}} e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{\infty} e^{i\omega[t_r - \vec{n}\cdot\vec{r}(t_r)/c]} \frac{d}{dt_r} \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{1 - \vec{\beta} \cdot \vec{n}} \right] dt_r \\ &= \frac{i\omega e}{\sqrt{4\pi c}} e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{\infty} [\vec{n} \times (\vec{n} \times \vec{\beta})] e^{i\omega[t_r - \vec{n}\cdot\vec{r}(t_r)/c]} dt_r\end{aligned}\quad (58)$$

To further the calculation, we need a specific particle trajectory. Consider a particle moving in an instantaneously circular orbit in the x - y plane. The center of the circle is at $(x, y, z) = (0, \rho, 0)$ and the particle is at $\vec{r}(0) = (0, 0, 0)$ at $t = 0$. The trajectory can be written as

$$\begin{cases} \vec{r}(t) = \rho \sin(\omega_B t) \vec{e}_x + \rho[1 - \cos(\omega_B t)] \vec{e}_y \\ \vec{\beta}(t) = \beta \cos(\omega_B t) \vec{e}_x + \beta \sin(\omega_B t) \vec{e}_y \end{cases}$$

where

$$\omega_B = c\beta/\rho$$

is the revolution frequency of the circular trajectory. Consider an arbitrarily located observation point with

$$\vec{n} = \cos \theta \cos \phi \vec{e}_x + \cos \theta \sin \phi \vec{e}_y + \sin \theta \vec{e}_z$$

where $\pi/2 - \theta$ is the angle between $\vec{n}$ and z axis and $(\theta, \phi) = (0, 0)$ is the direction along $\vec{\beta}(0)$. Then

$$\begin{aligned}\vec{n} \cdot \vec{r}(t) &= \rho \cos \theta \cos \phi \sin \omega_B t + \rho \cos \theta \sin \phi (1 - \cos \omega_B t) \\ &= \rho \cos \theta [\sin(\omega_B t - \phi) + \sin \phi]\end{aligned}$$

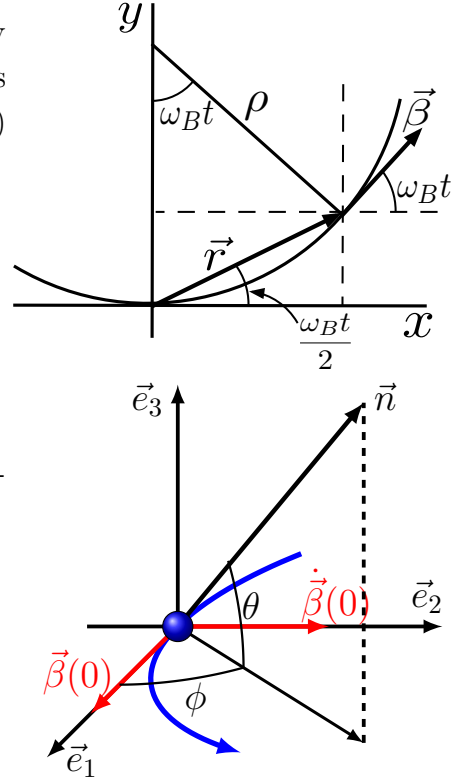
and the exponential function in the integral in Eq. (58), where we use t for t_r as the integration variable to simplify the notation, can be written as

$$\begin{aligned}e^{i\omega[t - \vec{n}\cdot\vec{r}(t)/c]} &= \exp \left\{ i \frac{\omega}{\omega_B} \left[\phi - \frac{\omega_B \rho}{c} \cos \theta \sin \phi \right] \right\} \\ &\times \exp \left\{ i \frac{\omega}{\omega_B} \left[(\omega_B t - \phi) - \frac{\omega_B \rho}{c} \cos \theta \sin(\omega_B t - \phi) \right] \right\} \\ &= \exp \left[i \frac{\omega}{\omega_B} (\phi - \mu \sin \phi) \right] \exp \left[i \frac{\omega}{\omega_B} (\alpha - \mu \sin \alpha) \right]\end{aligned}$$

where the integration variable t in Eq. (58) is changed to α with

$$\alpha = \omega_B t - \phi \quad \text{and} \quad \mu = (\omega_B \rho / c) \cos \theta = \beta \cos \theta$$

The term of vector cross-products in the integral in Eq. (58) can be calculated as



$$\begin{aligned}
\vec{n} \times [\vec{n} \times \vec{\beta}(t)] &= \vec{n}(\vec{n} \cdot \vec{\beta}) - \vec{\beta} = \beta \cos \theta \cos(\omega_B t - \phi) \vec{n} - \vec{\beta} \\
&= \beta [\cos^2 \theta \cos \phi \cos(\omega_B t - \phi) - \cos(\omega_B t)] \vec{e}_x \\
&+ \beta [\cos^2 \theta \sin \phi \cos(\omega_B t - \phi) - \sin(\omega_B t)] \vec{e}_y \\
&+ \beta \sin \theta \cos \theta \cos(\omega_B t - \phi) \vec{e}_z \\
&= \beta (\sin \phi \sin \alpha - \sin^2 \theta \cos \phi \cos \alpha) \vec{e}_x \\
&- \beta (\cos \phi \sin \alpha + \sin^2 \theta \sin \phi \cos \alpha) \vec{e}_y + \beta \sin \theta \cos \theta \cos \alpha \vec{e}_z
\end{aligned}$$

The Fourier transformation of $\vec{A}(t)$ can then be written as

$$\begin{aligned}
\vec{A}(\omega) &= \frac{i\omega e}{\omega_B \sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} e^{i(\omega/\omega_B)(\phi - \mu \sin \phi)} \int_{-\infty}^{\infty} [\vec{n} \times (\vec{n} \times \vec{\beta})] \exp \left[i \frac{\omega}{\omega_B} (\alpha - \mu \sin \alpha) \right] d\alpha \\
&= \frac{i\omega e \beta}{\omega_B \sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} e^{i(\omega/\omega_B)(\phi - \mu \sin \phi)} [(I_2 \sin \phi - I_1 \sin^2 \theta \cos \phi) \vec{e}_x \\
&\quad - (I_2 \cos \phi + I_1 \sin^2 \theta \sin \phi) \vec{e}_y + I_1 \sin \theta \cos \theta \vec{e}_z]
\end{aligned}$$

where

$$I_1 = \int_{-\infty}^{\infty} \exp \left[i \frac{\omega}{\omega_B} (\alpha - \mu \sin \alpha) \right] \cos \alpha d\alpha = 2 \int_0^{\infty} \cos \left[\frac{\omega}{\omega_B} (\alpha - \mu \sin \alpha) \right] \cos \alpha d\alpha \quad (59)$$

$$I_2 = \int_{-\infty}^{\infty} \exp \left[i \frac{\omega}{\omega_B} (\alpha - \mu \sin \alpha) \right] \sin \alpha d\alpha = 2i \int_0^{\infty} \sin \left[\frac{\omega}{\omega_B} (\alpha - \mu \sin \alpha) \right] \sin \alpha d\alpha \quad (60)$$

where $\mu = \beta \cos \theta$. For an ultra-relativistic particle, $\gamma \gg 1$, the radiation is concentrated in the radiation cone of $\theta \ll 1$ and $\phi \ll 1$ (see the coordinate in figures after Eq. (58))

$$\vec{A}(\omega) \stackrel{\gamma \gg 1}{\approx} \frac{i\omega e \beta}{\omega_B \sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} e^{i\nu(\phi - \mu \sin \phi)} (I_2 \sin \phi \vec{e}_x - I_2 \vec{e}_y + I_1 \sin \theta \vec{e}_z) \quad (61)$$

and

$$\begin{aligned}
|\vec{A}(\omega)|^2 &\stackrel{\gamma \gg 1}{\approx} \frac{e^2 \omega^2 \beta^2}{4\pi \omega_B^2 c} [|I_1|^2 \sin^2 \theta + |I_2|^2 (1 + \sin^2 \phi)] \\
&\stackrel{\sin^2 \phi \ll 1}{\approx} \frac{e^2 \omega^2 \beta^2}{4\pi \omega_B^2 c} (|I_1|^2 \sin^2 \theta + |I_2|^2) = |A_{\parallel}(\omega)|^2 + |A_{\perp}(\omega)|^2
\end{aligned} \quad (62)$$

where

$$|A_{\perp}(\omega)|^2 = \frac{e^2 \omega^2 \beta^2}{4\pi \omega_B^2 c} |I_1|^2 \sin^2 \theta \quad \text{and} \quad |A_{\parallel}(\omega)|^2 = \frac{e^2 \omega^2 \beta^2}{4\pi \omega_B^2 c} |I_2|^2$$

Note that $\theta = 0$ and $\phi = 0$ corresponds to $\vec{n} = \vec{e}_x$ that is parallel to $\vec{\beta}$ and the velocity of the particle at the retarded radiation time points toward the observation point. With $\gamma \gg 1$, inside the radiation cone at $\theta = 0$ and $\phi = 0$, the radiation spectrum is therefore

$$\frac{d^2 E}{d\omega d\Omega} = \frac{1}{\pi} |\vec{A}(\omega)|^2 = \frac{1}{\pi} |A_{\parallel}(\omega)|^2$$

Approximate I_1 and I_2 with Bessel Function for $\alpha \ll 1$

Since the radiation can only be observed in a very short duration of the radiation cone around $t = 0$ for the particle at $\vec{r} = 0$, the integrands of I_1 and I_2 are only significant when $\alpha = \omega_B t + \phi \ll 1$. Thus

$$\begin{aligned} \frac{I_1}{2} &= \int_0^\infty \cos \left[\frac{\omega}{\omega_B} (\alpha - \mu \sin \alpha) \right] \cos \alpha d\alpha \stackrel{\alpha \ll 1}{\simeq} \int_0^\infty \cos \left(\frac{\omega}{\omega_B} \left[(1 - \mu)\alpha + \frac{\mu}{3!} \alpha^3 \right] \right) d\alpha \\ \frac{I_2}{2i} &= \int_0^\infty \sin \left[\frac{\omega}{\omega_B} (\alpha - \mu \sin \alpha) \right] \sin \alpha d\alpha \stackrel{\alpha \ll 1}{\simeq} \int_0^\infty \alpha \sin \left(\frac{\omega}{\omega_B} \left[(1 - \mu)\alpha + \frac{\mu}{3!} \alpha^3 \right] \right) d\alpha \end{aligned}$$

where $\mu = \beta \cos \theta$. For $\gamma \gg 1$ and $\theta \ll 1$,

$$\mu = \beta \cos \theta \simeq \left(1 - \frac{1}{2\gamma^2} \right) \left(1 - \frac{1}{2}\theta^2 \right) = \frac{1}{4\gamma^2} (2\gamma^2 - 1)(2 - \theta^2) \simeq 1$$

and

$$1 - \mu \simeq 1 - \frac{1}{4\gamma^2} (2\gamma^2 - 1)(2 - \theta^2) = \frac{1}{4\gamma^2} [2 + (2\gamma^2 - 1)\theta^2] \simeq \frac{(1 + \gamma^2\theta^2)}{2\gamma^2}$$

Thus

$$\begin{aligned} \frac{I_1}{2} &\stackrel{\gamma \gg 1}{\simeq} \int_0^\infty \cos \left[\frac{\omega\alpha}{2\omega_B} \left(\frac{1 + \gamma^2\theta^2}{\gamma^2} + \frac{\alpha^2}{3} \right) \right] d\alpha \\ &= \left(\frac{1 + \gamma^2\theta^2}{\gamma^2} \right)^{1/2} \int_0^\infty \cos \left[\frac{3\xi}{2} \left(s + \frac{1}{3}s^3 \right) \right] ds = \left(\frac{1 + \gamma^2\theta^2}{\gamma^2} \right)^{1/2} \frac{1}{\sqrt{3}} K_{1/3}(\xi) \\ \frac{I_2}{2i} &\stackrel{\gamma \gg 1}{\simeq} \int_0^\infty \alpha \sin \left[\frac{\omega\alpha}{2\omega_B} \left(\frac{1 + \gamma^2\theta^2}{\gamma^2} + \frac{\alpha^2}{3} \right) \right] d\alpha \\ &= \left(\frac{1 + \gamma^2\theta^2}{\gamma^2} \right) \int_0^\infty s \sin \left[\frac{3\xi}{2} \left(s + \frac{1}{3}s^3 \right) \right] ds = \left(\frac{1 + \gamma^2\theta^2}{\gamma^2} \right) \frac{1}{\sqrt{3}} K_{2/3}(\xi) \end{aligned}$$

where the integration variable was changed from α to s as

$$\alpha = s \left(\frac{1 + \gamma^2\theta^2}{\gamma^2} \right)^{1/2}, \quad \xi = \frac{\omega}{3\omega_B} \left(\frac{1 + \gamma^2\theta^2}{\gamma^2} \right)^{3/2} \quad (63)$$

and $K_\nu(\xi)$ is the modified Bessel function with

$$K_{1/3}(\xi) = \sqrt{3} \int_0^\infty \cos \left[\frac{3\xi}{2} \left(s + \frac{1}{3}s^3 \right) \right] ds \quad (64)$$

$$K_{2/3}(\xi) = \sqrt{3} \int_0^\infty s \sin \left[\frac{3\xi}{2} \left(s + \frac{1}{3}s^3 \right) \right] ds \quad (65)$$

Spectrum of Synchrotron Radiation

$$\begin{aligned}
\frac{d^2 E}{d\omega d\Omega} &= \frac{1}{\pi} |\vec{A}(\omega)|^2 = \frac{1}{c} \left(\frac{e\omega\beta}{2\pi\omega_B} \right)^2 (|I_2|^2 + |I_1|^2 \sin^2 \theta) \\
&= \frac{1}{3c} \left(\frac{e\omega\beta}{\pi\omega_B} \right)^2 \left(\frac{1 + \gamma^2 \theta^2}{\gamma^2} \right)^2 \left[K_{2/3}^2(\xi) + \left(\frac{\gamma^2 \sin^2 \theta}{1 + \gamma^2 \theta^2} \right) K_{1/3}^2(\xi) \right] \\
&= \frac{3}{c} \left(\frac{e\beta}{\pi} \right)^2 \frac{\gamma^2}{1 + \gamma^2 \theta^2} \left[\xi^2 K_{2/3}^2(\xi) + \left(\frac{\gamma^2 \theta^2}{1 + \gamma^2 \theta^2} \right) \xi^2 K_{1/3}^2(\xi) \right] \quad (66)
\end{aligned}$$

As shown in the figure, the maximal radiation is at $\xi \sim 1/2$. A critical frequency ω_c of the radiation is defined as the frequency for $\xi = 1$ at $\theta = 0$. From ξ in Eq. (63),

$$\xi \stackrel{\theta=0}{=} 1 \implies \omega_c = 3\gamma^3 \omega_B = \frac{3\gamma^3 v}{\rho}$$

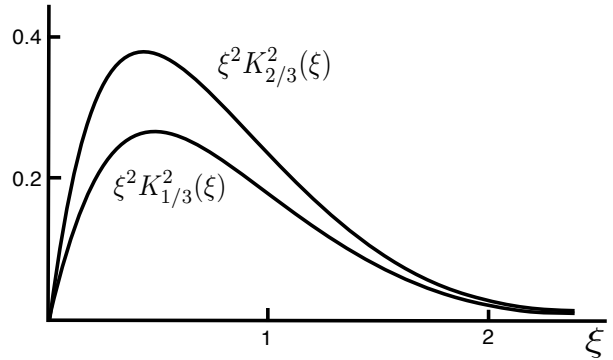
where $\omega_B = v/\rho$. With the definition of the critical frequency,

$$\xi(\theta) = \frac{\omega}{\omega_c} (1 + \gamma^2 \theta^2)^{3/2}$$

For $\omega > \omega_c$, the radiation vanishes exponentially. For the Advanced Photon Source in Argonne, for example,

$$\left\{ \begin{array}{l} \gamma = \frac{7 \text{ GeV}}{0.5 \text{ MeV}} = 14000 \\ \rho = \frac{1100 \text{ m}}{2\pi} = 175 \text{ m} \end{array} \right. \implies \left\{ \begin{array}{l} \omega_B = \frac{c}{\rho} \simeq 1.7 \times 10^6 \text{ Hz} \\ \omega_c \simeq 3\gamma^3 \omega_B \simeq 1.4 \times 10^{19} \text{ Hz} \end{array} \right.$$

where ω_c is in the range of hard X-ray. Note that the peak of the radiation spectrum locates roughly at $\xi \simeq 1/2$ that corresponds to $\omega \simeq \omega_c/2$.



Angle Spread θ_c of Radiation Cone

Since when $\xi(\theta) > 2$ there is very little radiation and

$$\xi(\theta) = \frac{\omega}{\omega_c} (1 + \gamma^2 \theta^2)^{3/2} \quad (67)$$

depends both on ω and θ , the width of the radiation cone depends also on the frequency ω of the radiation for a given γ . We now study the frequency-dependence of the radiation-cone width. For a given frequency of $\omega < \omega_c$, the center of the radiation cone is at $\theta = 0$ that corresponds to

$$\xi = \xi(0) = \frac{\omega}{\omega_B} \quad \text{for } \theta = 0$$

When $\xi(\theta) > 2$ with $\omega < \omega_c$, the radiation vanishes exponentially. We can thus define the edge of the radiation cone, $\theta = \theta_c$, is at

$$\xi(\theta_c) \simeq \xi(0) + 2 = \frac{\omega}{\omega_c} + 2$$

which yields

$$\frac{\omega}{\omega_c} (1 + \gamma^2 \theta_c^2)^{3/2} \simeq \frac{\omega}{\omega_c} + 2 \quad \Rightarrow \quad \theta_c \simeq \frac{1}{\gamma} \sqrt{\left(1 + \frac{2\omega_c}{\omega}\right)^{2/3} - 1}$$

The total radiation energy from the radiation cone can be estimated as

$$\frac{dE}{d\omega} \sim \left. \frac{d^2 E}{d\omega d\Omega} \right|_{\theta=0} (2\pi\theta_c)$$

Total Radiation Energy Per Solid Angle

With the energy density of the radiation calculated in Eq. (66), the total energy can be calculated as

$$\begin{aligned} \frac{dE}{d\Omega} &= \int_0^\infty \frac{d^2 E}{d\omega d\Omega} d\omega = \frac{\omega_c}{(1 + \gamma^2 \theta^2)^{3/2}} \int_0^\infty \frac{d^2 E}{d\omega d\Omega} d\xi \\ &= \left(\frac{3e^2}{\pi^2 c} \right) \frac{\gamma^2 \omega_c}{(1 + \gamma^2 \theta^2)^{5/2}} \left[\int_0^\infty \xi^2 K_{2/3}^2(\xi) d\xi + \frac{\gamma^2 \theta^2}{1 + \gamma^2 \theta^2} \int_0^\infty \xi^2 K_{1/3}^2(\xi) d\xi \right] \\ &= \left(\frac{3e^2}{\pi^2 c} \right) \left(C_2 + C_1 \frac{\gamma^2 \theta^2}{1 + \gamma^2 \theta^2} \right) \frac{\gamma^2 \omega_c}{(1 + \gamma^2 \theta^2)^{5/2}} \end{aligned}$$

where C_1 and C_2 are two numerical constants,

$$C_1 = \int_0^\infty \xi^2 K_{1/3}^2(\xi) d\xi \quad \text{and} \quad C_2 = \int_0^\infty \xi^2 K_{2/3}^2(\xi) d\xi$$

14.7 Wigglers and Undulators for Synchrotron Light Sources

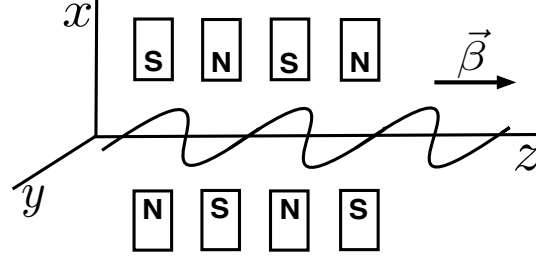
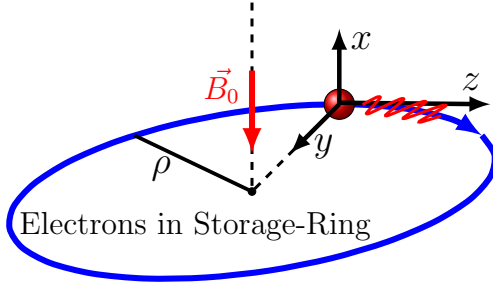
For synchrotron light sources, in order to achieve a broad band spectrum of the radiation, one needs to increase the critical frequency

$$\omega_c = \frac{3\gamma^3 v}{\rho}$$

For a given beam energy with a fixed γ , only way to increase ω_c is to reduce ρ that is the instantaneous radius of curvature of beam trajectory. With a fixed beam energy, the radiation power calculated in Eq. (46)

$$P(t_r) = \frac{2e^2 c}{3\rho^2} \gamma^4 \beta^4$$

also increases quadratically with the decrease of ρ . Wigglers and undulators were thus developed to drive an ultra relativistic particle beam into a very tightly periodically banded trajectory. Inside a wiggler or undulator, the charge particles oscillate in the transverse plane



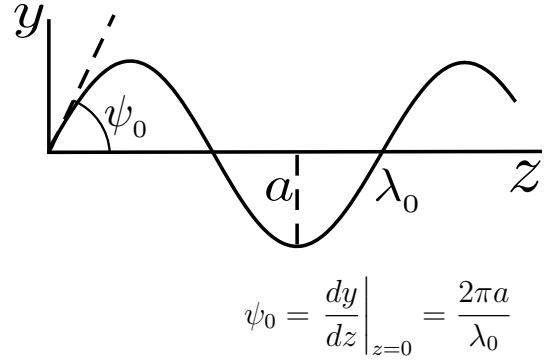
$$F_y = ev_z(B_0 - B_x), \quad \text{with } |B_x| \ll B_0$$

as the beam travels through the wiggler/undulator with near the speed of light in the longitudinal direction. The trajectory of beam particles can be approximated as

$$y \simeq a \sin\left(\frac{2\pi z}{\lambda_0}\right) \quad \text{and} \quad z = c\beta t$$

where λ_0 is the period of the magnet structure of the wiggler/undulator. The maximal transverse speed of the beam particles is

$$v = \frac{dy}{dt} = \frac{dy}{dz} \bigg|_{z=0} \frac{dz}{dt} = \frac{2\pi a}{\lambda_0} c\beta \simeq \psi_0 c$$



The radius of curvature of the trajectory is

$$\rho^{-1} = \frac{|d^2y/dz^2|}{[1 + (dy/dz)^2]^{3/2}} \simeq \left| \frac{d^2y}{dz^2} \right| = a \left(\frac{2\pi}{\lambda_0} \right)^2 \left| \sin\left(\frac{2\pi z}{\lambda_0}\right) \right|$$

where $dy/dz \leq \psi_0 \ll 1$, and the minimal radius of curvature is

$$\rho_{min} = (\rho^{-1})_{max} = \frac{\lambda_0^2}{4\pi^2 a} = \frac{\lambda_0}{2\pi\psi_0}$$

The maximal critical frequency of the synchrotron radiation of the beam inside the wiggler/undulator is thus

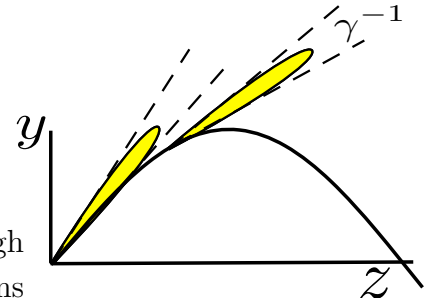
$$\omega_c = \frac{3\gamma^3 v}{\rho_{min}} \sim \gamma^3 v \frac{\psi_0}{\lambda_0} \quad \Rightarrow \quad \text{broad spectrum of radiation}$$

a) Wiggler

$$\psi_0 \gg \frac{1}{\gamma} \quad \text{or} \quad \frac{\lambda_0}{a} \ll \gamma$$

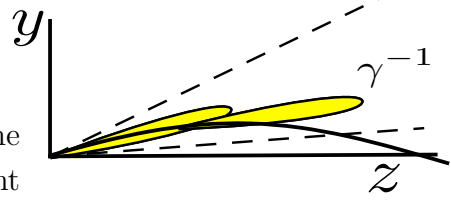
In this case, the wiggling of beam trajectory is larger enough so that the radiation cones of the beam at two nearby positions direct at different direction and are not overlapped for an observer at a fixed location. If the magnet structure of a wiggler has N periods, the intensity of the light I is the incoherent addition of the radiation strength I_1 from a single period of the bending,

$$I = NI_1$$



b) Undulators

$$\psi_0 \ll \frac{1}{\gamma}$$



In this case, many radiation cones are overlapped and the radiation field observed at a fixed location is the coherent superposition of the continuously radiated field along the oscillation of the trajectory. Because of the coherent superposition, the field with certain wavelengths could be in constructive addition and the others in destructive addition. Consequently, there are only one or a few peaks in the frequency spectrum of the radiation.

14.8 Thomson Scattering of Radiation

When a plane electromagnetic wave is shined on a charged particle, the particle will be accelerated and in turn emit radiation. This process can also be described as scattering of an incident electromagnetic wave. The electric field of a plane wave of monochromatic electromagnetic incident wave can be written as

$$\vec{E}_{in}(\vec{x}, t) = \vec{\epsilon}_0 E_0 e^{i(\vec{k}_0 \cdot \vec{r} - \omega t)} \implies \vec{\epsilon}_0 E_0 \cos(\vec{k}_0 \cdot \vec{r} - \omega t) \quad (68)$$

where the wave propagates in the direction of $\vec{k}$ and $\vec{E}$ is polarized in the direction of $\vec{\epsilon}_0$ with $\vec{\epsilon}_0 \cdot \vec{\epsilon}_0 = 1$ and $\vec{\epsilon}_0 \cdot \vec{k}_0 = 0$. The acceleration of a non-relativistic charged particle inside the electromagnetic wave is

$$\dot{\vec{v}} = \frac{e}{m} \vec{E}_{in} = \vec{\epsilon}_0 \frac{eE_0}{m} \cos(\vec{k}_0 \cdot \vec{r} - \omega t) \quad (69)$$

where the magnetic force from the electromagnetic field is negligible for a non-relativistic motion. The radiation emitted by this accelerated non-relativistic charged particle is then

$$\begin{aligned} \frac{dP(t)}{d\Omega} &= \frac{e^2}{4\pi c} \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{n} \cdot \vec{\beta})^5} \stackrel{\beta \ll 1}{\simeq} \frac{e^2}{4\pi c^3} \left| \vec{n} \times (\vec{n} \times \dot{\vec{v}}) \right|^2 \\ &= \frac{c}{4\pi} \left(\frac{e^2}{mc^2} \right)^2 |E_0|^2 |\vec{\epsilon}_0 - (\vec{n} \cdot \vec{\epsilon}_0)\vec{n}|^2 \cos^2(\vec{k}_0 \cdot \vec{r}(t) - \omega t) \end{aligned} \quad (70)$$

where $\vec{\epsilon}_0 - (\vec{n} \cdot \vec{\epsilon}_0)\vec{n}$ is the radiation field polarized in the transverse plane with $\vec{n}$ as the normal. Note that $\vec{\epsilon}_0 - (\vec{n} \cdot \vec{\epsilon}_0)\vec{n} = 0$ when $\vec{n} \parallel \vec{\epsilon}_0$.

Average Radiation Power in One Wave Oscillation

Consider the case that during one period $T = 2\pi/\omega$ of the electromagnetic wave, the particle moves a negligible distance as compared with the wavelength $2\pi/k_0$ of the electromagnetic wave. This is generally true with a non-relativistic particle. Therefore

$$\frac{1}{T} \int_0^T \cos^2(\vec{k}_0 \cdot \vec{r}(t') - \omega t) dt \simeq \frac{1}{T} \int_0^T \cos^2(\text{constant} - \omega t) dt \stackrel{s=\omega t}{=} \frac{1}{2\pi} \int_0^{2\pi} \cos^2 s ds = \frac{1}{2}$$

and

$$\left\langle \frac{dP}{d\Omega} \right\rangle = \frac{c}{8\pi} \left(\frac{e^2}{mc^2} \right)^2 |E_0|^2 |\vec{\epsilon}_0 - (\vec{n} \cdot \vec{\epsilon}_0) \vec{n}|^2 \quad (71)$$

Scattering Cross Section

This wave-particle interaction can also be viewed as a wave scattered by a free charged particle. The differential scattering cross section is defined as

$$\frac{d\sigma}{d\Omega} = \frac{\text{Energy radiated per unit time and per unit solid angle}}{\text{Energy in incident wave per unit area and per unit time}}$$

where

$$\begin{aligned} \text{Energy radiated per unit time and per unit solid angle} &= \left\langle \frac{dP}{d\Omega} \right\rangle \\ \text{Energy in incident wave per unit area and per unit time} &= \frac{c}{4\pi} \langle E^2 \rangle = \frac{c}{8\pi} E_0^2 \end{aligned}$$

Thus

$$\frac{d\sigma}{d\Omega} = \left(\frac{e^2}{mc^2} \right)^2 |\vec{\epsilon}_0 - (\vec{n} \cdot \vec{\epsilon}_0) \vec{n}|^2 = \left(\frac{e^2}{mc^2} \right)^2 [1 - (\vec{n} \cdot \vec{\epsilon}_0)^2] \quad (72)$$

Note that $\vec{\epsilon}_0 - (\vec{n} \cdot \vec{\epsilon}_0) \vec{n}$ is the radiation field polarized in the transverse plane that has $\vec{n}$ as its normal direction. In the Jackson's book, this transverse polarization is parameterized by two perpendicular unit vectors $(\vec{\epsilon}_1, \vec{\epsilon}_2)$ on the transverse plane, *i.e.*

$$(\vec{\epsilon}_1, \vec{\epsilon}_2) = \vec{\epsilon}_0 - (\vec{n} \cdot \vec{\epsilon}_0) \vec{n}$$

where $\vec{\epsilon}_1 \cdot \vec{\epsilon}_2 = 0$.

For Unpolarized Incident Wave

To calculate $(\vec{n} \cdot \vec{\epsilon}_0)$ in Eq. (72), we need to choose a coordinate. Let the incident wave travel in the z direction and $\vec{\epsilon}_0$ be in the $x - y$ plane. If the incident wave is unpolarized,

$$\vec{\epsilon}_0 = \frac{1}{\sqrt{2}}(1, 1, 0)$$

and $\vec{n}$ is in an arbitrary direction,

$$\vec{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

where θ is the angle between $\vec{n}$ and $\vec{k}_0$. Then

$$(\vec{n} \cdot \vec{\epsilon}_0)^2 = \frac{1}{2}(\sin \theta \cos \phi + \sin \theta \sin \phi)^2 = \frac{1}{2} \sin^2 \theta (1 + \sin 2\phi)$$

and

$$\frac{d\sigma}{d\Omega} = \frac{1}{2} \left(\frac{e^2}{mc^2} \right)^2 (1 + \cos^2 \theta - \sin^2 \theta \sin 2\phi)$$

The total scattering cross section is

$$\sigma = \frac{1}{2} \left(\frac{e^2}{mc^2} \right)^2 \int (1 + \cos^2 \theta - \sin^2 \theta \sin 2\phi) \sin \theta d\theta d\phi = \frac{8\pi}{3} \left(\frac{e^2}{mc^2} \right)^2 \quad (73)$$

This is called Thomson cross section.

Frequency of Scattered Wave

The radiated field is

$$\vec{E}_{rad}(\vec{x}, t) = \frac{e}{c} \left\{ \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{R(1 - \vec{n} \cdot \vec{\beta})^3} \right\}_{retarded} \simeq \frac{e^2 E_0}{mc} \left[\frac{\vec{n} \times (\vec{n} \times \vec{\epsilon}_0)}{R} \right] e^{i[\vec{k}_0 \cdot \vec{r}(t') - \omega t']} \quad (74)$$

where $t' = t - R(t')/c$. During each oscillation of incident wave, a non-relativistic charged particle moves very little and, therefore, R and $\vec{n}$ can be treated as constants. In an ordinary lab scale, moreover, $R/c \ll 1/\omega$ and $t \simeq t'$. Thus

$$\vec{E}_{rad}(\vec{x}, t) \sim e^{i(\vec{k}_0 \cdot \vec{r} - \omega t)}$$

i.e. the radiated or scattered wave has the same frequency as that of the incident wave.

Quantum Correction

If the mass of the charged particle is so small that the photon momentum $\hbar\omega/c$ of the incident wave is comparable to or larger than mc , the momentum (or energy) change of the scattered photon due to the photon-particle collision has to be considered. Including the effect of elastic collision between an photon and the particle, the Thomson differential scattering cross section in Eq. (72) is modified as

$$\frac{d\sigma}{d\Omega} = \left(\frac{e^2}{mc^2} \right)^2 \left(\frac{k}{k_0} \right)^2 [1 - (\vec{n} \cdot \vec{\epsilon}_0)^2]$$

where $k_0 = \omega_0/c$ and $k = \omega/c$ are the wave number of the incident and scattered wave, respectively, and $k < k_0$ due to the recoil of the particle during the photon-particle collision.

Summary of Radiation Field By Moving Charge

Total field emitted by a moving charge

$$\begin{aligned}\vec{E}(\vec{x}, t) &= q \left\{ \frac{\vec{n} - \vec{\beta}}{\gamma^2 R^2 (1 - \vec{n} \cdot \vec{\beta})^3} + \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{c R (1 - \vec{n} \cdot \vec{\beta})^3} \right\}_{ret} \\ \vec{B}(\vec{x}, t) &= \left(\vec{n} \times \vec{E} \right)_{ret}\end{aligned}$$

The radiation field is

$$\vec{E}_{rad}(\vec{x}, t) = q \left. \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{c R (1 - \vec{n} \cdot \vec{\beta})^3} \right|_{ret}$$

Polarization of the Field

$\vec{B}$ is always perpendicular to the direction of the propagation direction $\vec{n}$, *i.e.* $\vec{B} \cdot \vec{n} = 0$.

The static part of $\vec{E}$ is along the direction of $\vec{n}$, *i.e.*

$$\vec{n} \cdot \vec{E} = \frac{q}{\gamma^2 R^2 (1 - \vec{n} \cdot \vec{\beta})^2}$$

$\vec{E}_{rad}$ is perpendicular to both $\vec{n}$ and $\vec{B}$, $\vec{E}_{rad} \cdot \vec{n} = 0$ and $\vec{B} \cdot \vec{E} = 0$.

Radiation Power Per Solid Angle

$$\left(\frac{dP}{d\Omega} \right)_{rad} = \frac{c}{4\pi} R^2 |\vec{E}_{rad}|^2 = \frac{c}{4\pi} R^2 |\vec{B}|^2$$

Note that the energy radiated through a unit solid angle is independent of R , the distance to the observer, *i.e.* the total radiation energy on a sphere that is centered at the radiation source does not decay with the radius R .

Frequency Spectrum of Radiation Energy

We can use either

$$\vec{A}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\sqrt{\frac{c}{4\pi}} R \vec{E}_{rad}(\vec{r}, t) \right] e^{i\omega t} dt \quad \text{or} \quad \vec{C}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\sqrt{\frac{c}{4\pi}} R \vec{B}(\vec{r}, t) \right] e^{i\omega t} dt$$

Then

$$\left(\frac{dW}{d\Omega d\omega} \right)_{rad} = 2 |\vec{A}(\omega)|^2 = 2 |\vec{C}(\omega)|^2$$

For $|\vec{R}| \gg |\vec{r}|$, $\vec{n} \simeq \text{constant}$ and

$$\begin{aligned}\vec{A}(\omega) &= i\omega \left(\frac{e^2}{8\pi^2 c} \right)^{1/2} e^{i\omega(\vec{x} \cdot \vec{n})/c} \int_{-\infty}^{\infty} \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] e^{i\omega[t - \vec{n} \cdot \vec{r}(t)/c]} dt \\ \vec{C}(\omega) &= \vec{n} \times \vec{A}(\omega) = i\omega \left(\frac{e^2}{8\pi^2 c} \right)^{1/2} e^{i\omega(\vec{x} \cdot \vec{n})/c} \int_{-\infty}^{\infty} (\vec{n} \times \vec{\beta}) e^{i\omega[t - \vec{n} \cdot \vec{r}(t)/c]} dt\end{aligned}$$

Another way to calculate the radiation spectrum

To further the calculation, we consider an instantaneously circular orbit in the $y - z$ plane. The center of the circular orbit is at $(x, y, z) = (0, \rho, 0)$ and the particle is at $\vec{r}(0) = (0, 0, 0)$ at $t = 0$. The particle orbit can be written as

$$\begin{cases} \vec{r}(t) = \rho[1 - \cos(\omega_B t)] \vec{e}_y + \rho \sin(\omega_B t) \vec{e}_z \\ \vec{\beta}(t) = \beta \sin(\omega_B t) \vec{e}_y + \beta \cos(\omega_B t) \vec{e}_z \end{cases}$$

where $\omega_B = c\beta/\rho$ is the revolution frequency of the particle. Consider the radiation occurs when the particle is around $\vec{r} = \vec{r}(0) = 0$,

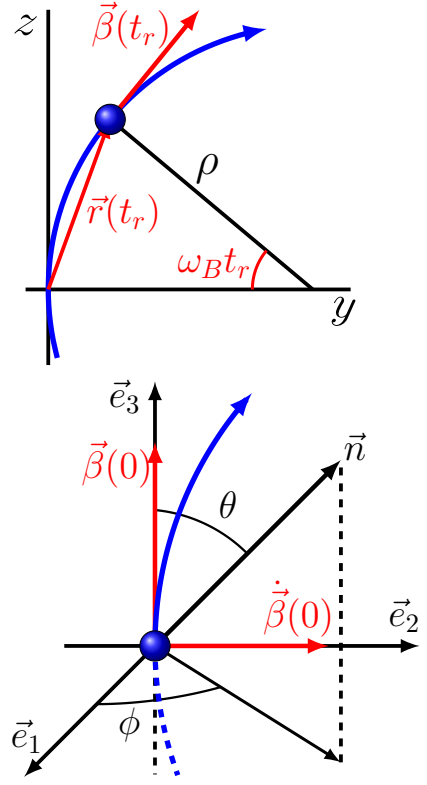
$$\vec{n} = \frac{\vec{x} - \vec{r}(t_r)}{|\vec{x} - \vec{r}(t_r)|} \simeq \frac{\vec{x}}{|\vec{x}|}$$

For an arbitrarily observation point $\vec{x}$, we can write

$$\vec{n} = \sin \theta \cos \phi \vec{e}_x + \sin \theta \sin \phi \vec{e}_y + \cos \theta \vec{e}_z$$

where θ is the angle from the observation direction, and

$$\vec{n} \cdot \vec{r}(t) = \rho (\sin \theta \sin \phi + \cos \theta \sin \omega_B t - \sin \theta \sin \phi \cos \omega_B t)$$



Assignment 14-1. 14.1, 14.4

Assignment 14-2. 14.5, 14.7, 14.9

Assignment 14-3. 14.10, 14.12, 14.14

Assignment 14-4. 14.26

HW Assignment 14.1, 14.3, 14.4, 14.5, 14.7, 14.9, 14.10, 14.11, 14.12, 14.14, 14.26